Language Modeling as Equilibrium Computation: Weight-Tied Depth, Magnetic Dynamics, and Truthful Token Auctions

Sharath Sathish
https://github.com/SharathSPhD/game-llm

August 2026

Abstract

Language modeling is reformulated as equilibrium computation across three stages: training dynamics via Magnetic Mirror Descent toward quantal response equilibria, model depth as a weight-tied transformer block solved to a fixed point (EqLM), and multi-model decoding through truthful second-price token auctions. Six hypotheses carrying pre-registered quantitative gates were empirically adjudicated, with every finding subjected to adversarial review. An anytime-unrolled training regime closes the width gap to achieve a weight-tied single-block language model that reaches a ratio of 0.991 of a parameter-matched 12-layer explicit transformer's BLiMP at 121M parameters, trains 2.1 times faster, and gracefully trades off inference cost through a dial over budgets $\{4, 8, 12\}$ iterations. A trajectory-local contraction penalty yields the first certified 121M-parameter equilibrium (convergence rate 1.0 at 4.0 mean iterations), resolving the gap between quality and certification into two separately solved halves whose naive combination is provably refuted by its own pre-registered prediction. On matrix games, Magnetic Mirror Descent converges linearly (log-linear R^2 of 0.995 on matching pennies) where simultaneous gradient play cycles; in preference optimization, the magnetic anchor proves second-order to the preference gradient across four orders of magnitude of regularization strength. Truthful second-price token auctions over domain specialists beat the best single specialist by 23% and uniform logit averaging by 12% on mixed-domain perplexity at scoring time, yet this advantage inverts in closed-loop generation, an inversion the adversarial reviewer correctly forced into scope before the follow-up experiment demonstrated its necessity. The program closes with two formally missed hypotheses and identifies quality-preserving certification as a sharply posed open problem.

1 Introduction

Contemporary language models are trained by minimizing a single loss with a single optimizer, a regime one may characterize as dictatorial optimization. Yet the systems these models inhabit are interactive: alignment tunes a policy against a frozen reference; decoding aggregates competing continuations; and training itself is increasingly a game between model, data, and reference. Game theory offers the natural mathematics for such settings, transforming from side tools to core methodology.

Interactive settings demand game-theoretic thinking. A language model in deployment does not optimize in isolation; it inherits a loss landscape shaped by its training data, its reference model (in alignment), and external judges or users whose preferences it must anticipate. Each of these interacting agents—model, data curator, reference—follows its own gradient, and the resulting dynamics are rarely convex. The synchronized gradient-descent-ascent dynamics that emerge from adversarial training are known to cycle (Mertikopoulos et al., 2018), motivating the search for alternative equilibrium-seeking algorithms. Decoding, too, is a game: when selecting the next token, the model competes against its own continuation distribution and the hidden regularities in test-time data. Mechanism design enters when aggregating multiple models: a weighted ensemble implicitly assumes the aggregator has reliable information about each model's quality, but in truth that quality is adversarially determined by each model's incentive to maximize its selection probability.

Prior work has begun importing game-theoretic machinery into language modeling, importing both techniques and theory. Mirror-descent methods with last-iterate convergence guarantees in zero-sum games (Sokota et al., 2023) offer principled training dynamics where simultaneous play converges. Quantal response equilibrium (McKelvey, 1995) connects inverse-temperature rationality (softmax temperature) to Nash equilibrium behavior, bridging game theory and modern decoding. Deep-equilibrium models (Bai et al., 2019; Dupont et al., 2021) implement computation as fixed-point solve rather than feed-forward pass, achieving $O(1)$ depth-memory complexity and opening a pathway to equilibrium-based inference. Mechanism design (Duetting et al., 2024) frames model aggregation as a truthful auction where each model is a bidder, with Vickrey's second-price auction guaranteeing incentive compatibility. Yet these strands remain scattered: no end-to-end language modeling system has integrated training dynamics (mirror descent), model depth (DEQ), and inference aggregation (auctions) simultaneously, nor measured whether the integrated system survives empirical thresholds set before experiments begin.

Prior work left three open questions. First: does mirror descent's theoretical convergence (last-iterate linear convergence in zero-sum games) produce better training dynamics for language models in practice, or does the benefit evaporate at realistic scale? Second: can weight-tied deep-equilibrium architectures match explicit stacks in language modeling quality, and if not, where does the gap come from—is it a fundamental architectural limit or a training-regime mismatch? Third: can game-theoretic mechanism design (truthful auctions) improve multi-model inference without requiring external judges or reward signals to calibrate weights? None of these questions have satisfactory answers in the published literature at small-LM scale with pre-registered hypotheses and adversarial review.

This paper asks a blunt empirical question: *when the equilibrium view is implemented end-to-end at small-LM scale, what survives contact with pre-registered thresholds?* A single codebase (`kinetic_ai`) was built implementing Magnetic Mirror Descent, quantal response equilibrium solvers, deep-equilibrium language models, truthful token auctions, and parameter-space magnetic anchoring in preference optimization; six hypotheses were registered with quantitative pass/fail gates before their experiments ran; and every finding underwent an adversarial integrity review (Tarka) empowered to refute, rescope, or demand re-analysis. The review materially overturned the initial conclusions twice: it refuted the proposed mechanistic explanation for the scaling gap in solver-based EqLM training (Section 5.4), and it forced a scoring-time scoping onto the auction result that a subsequent experiment proved necessary (Section 5.8). The full diagnostic arc is reported here, including two formally missed hypotheses, on the principle that a well-documented null carries as much design information as a positive result.

The contributions of this work are as follows. An equilibrium language model (Figure 1) reaches parity with a parameter-matched 12-layer explicit transformer at 121M parameters when trained with anytime supervision: a weight-tied single block trained unrolled with cross-entropy supervision on intermediate iterates z_4, z_8, z_{12} achieves BLiMP ratio 0.991 (bootstrap 95% CI $[0.990, 0.992]$, three seeds) and trains 2.1× faster than training via implicit differentiation, with graceful degradation across inference budgets $\{4, 8, 12\}$ from one checkpoint (F24). A certified 121M-parameter equilibrium is separately achieved: a trajectory-local finite-difference contraction penalty produces solver convergence rate 1.0 at 4.0 mean iterations and the lowest peak memory of any arm, though at a quality cost; the naive combination of anytime supervision and the penalty is refuted by its own pre-registered prediction, leaving quality-preserving certification as a sharply posed open problem (F24, B1–B4). Convergence of training dynamics is proven on matrix games: Magnetic Mirror Descent converges linearly (log-linear $R^2 = 0.99$) where simultaneous gradient ascent-descent cycles, periodic reference resets recover Nash equilibrium to NashConv $< 10^{-5}$, and an asymmetric-game discovery shows that uniform-anchor fixed points diverge from logit-QRE in the general case (F1–F3). The parameter-space magnetic anchor in preference optimization is real but second-order: at the pre-registered regularization strengths the held-out accuracy matches DPO exactly, and a dose-response sweep over four orders of magnitude ($\tau \in [10^{-3}, 10]$) fails to produce the predicted reduction in reward-hacking drift, leaving the magnetic pull as a pretraining-phase tool rather than a preference-phase one (F21). Auction aggregation is correctly scoped (Figure 2): truthful second-price token auctions beat the best single domain specialist by 23% on mixed-domain perplexity at scoring time (F22: auction 182.8 ppl vs. best single 236.8 ppl), yet lose in closed-loop generation where the independent judge's own style prior dominates between-system effects (F23: judge NLL 4.74 vs. best single 3.39, judge-relative result) (F22–F23). Finally, a conversion-damage measurement on Qwen3-1.7B prior to uptraining identifies which layers can be safely tied: average layer initialization preserves far more function than stepwise (1900 vs. 220,000 ppl at 8+8 outer layers), outer layers matter more than the number of core iterations (10-fold improvement adding two explicit layers vs. 1% improvement adding a second core), and all configurations start heavily damaged (300–3000× base perplexity), pinning uptraining budget rather than surgery as the binding constraint (F25).

The empirical approach taken here—pre-registration of quantitative gates, independent recomputation of every reported quantity, and equal standing for negative outcomes—grounds these theoretical ideas in evidence at realistic scale. The hypotheses test whether game-theoretic principles, when implemented faithfully, improve language model training and inference. The measurement framework enforces not just statistical significance but pre-registered thresholds, chosen before experiments ran to avoid post-hoc moving of goalposts. The adversarial review process (Tarka) acts as a built-in skeptical reader, empowered to refute claims and demand evidence. Two of the six hypotheses (H1 and H5) formally missed their gates and two (H3 and H6) returned partial outcomes; those results are reported at the same length and with the same evidentiary standard as the outcomes that passed. This methodology constrains the narrative: strong claims require matching pre-registered thresholds, and open problems are posed sharply rather than smoothed over.

The paper proceeds as follows. Section 5.1 establishes the convergence properties of Magnetic Mirror Descent on matrix games and derives the QRE solver engineering improvements. Section 5.2 validates the truthfulness of second-price token auctions empirically. Section 5.3 reports the full diagnostic arc of EqLM development from initialization bugs through solver-awareness and into anytime training, ending with the parity result and the open problem of quality-preserving certification. Section 5.7 measures the parameter-space magnetic anchor's contribution to preference optimization and documents the drift-resistance property of the equilibrium architecture. Section 5.8 reports auction decoding at scoring time and its failure in closed-loop generation, with the metric pathology that explains the miss. Section 5.9 describes the uptraining pathway to realistic-scale models, reports the conversion-damage curve, and identifies the operating point for the follow-on experiments. The discussion synthesizes findings across the program, identifies the quality-preserving certification challenge as the sharpest remaining open problem, and discusses implications for practitioners converting pretrained models to recursive topologies. Reproducibility is prioritized throughout: every number traces to a results file with its config hash and git commit, adversarial review verdicts are documented for each finding, and pre-registration gates are stated explicitly with their outcomes.

2 Background and Related Work

Contemporary language model optimization is dominated by a single loss function and a single optimizer. One might caricature this regime as *dictatorial optimization*. Yet the systems in which these models operate are inherently interactive: preference tuning optimizes a policy against a frozen reference, decoding aggregates diverse model outputs, and training increasingly unfolds as a feedback loop between model, data, and reference. Game-theoretic machinery offers the natural mathematical language for such settings. A growing line of recent work has begun to import equilibrium theory and mechanism design into language modeling [Sokota et al., 2023, Bai et al., 2019, Dütting et al., 2024, Wu et al., 2024]. This section reviews the foundational concepts and prior work that motivate the equilibrium-theoretic treatment of three stages of language modeling: training dynamics via magnetic mirror descent, model depth via implicit fixed-point computation, and decoding via truthful mechanism design.

2.1 Quantal Response Equilibrium and Softmax as Bounded Rationality

Quantal response equilibrium (QRE), introduced by McKelvey and Palfrey in a foundational 1995 paper on normal-form games, models boundedly rational agents by replacing best-response play with stochastic best response [McKelvey and Palfrey, 1995]. Under a QRE, an agent plays action a with probability proportional to $\exp(\lambda\, u(a))$, where $u(a)$ is the payoff and λ is a rationality (inverse-temperature) parameter that captures the agent's degree of strategic optimization. As $\lambda \to \infty$, the QRE converges to Nash equilibrium; at intermediate λ it interpolates between uniform random play (low λ) and best response (high λ), and coincides with entropy-regularized equilibrium. The normalizing constant, $\sum_a \exp(\lambda\, u(a))$, is the softmax partition function. QRE has been widely studied in experimental economics and behavioral game theory to characterize how real players deviate from perfect rationality.

The connection to language modeling is direct and structural. The softmax output layer of any modern language model is literally a quantal response to its logits. At each token position, the model computes a vector of scores (logits $\ell \in \mathbb{R}^V$) and outputs a probability distribution via softmax: $p_i = \exp(\ell_i)/Z$, where $Z = \sum_j \exp(\ell_j)$ is the partition function. The temperature parameter $\tau = 1/\lambda$ controls the peakedness of this distribution. Low temperatures (high λ) concentrate probability sharply on high-scoring tokens; high temperatures spread probability more evenly across continuations. This observation, though elementary, motivates a systematic study of language model decoding through the lens of QRE: the model's token distribution traces out QRE paths as temperature varies, and one may interrogate what equilibrium properties the learned logits satisfy. Such paths arise naturally in games where agents adapt their strategies in response to fixed payoff structures. Understanding whether language models' learned representations align with QRE predictions is an open empirical question and is explored in Section 5.1 of the present work.

2.2 Mirror Descent, Optimization Geometry, and Regularized Learning

Mirror descent, a classical technique in convex optimization, generalizes gradient descent by replacing the Euclidean projection step with a projection in a non-Euclidean geometry defined by a *mirror map* (or Bregman divergence). When applied to probability distributions on a simplex, using the negative entropy as the mirror map naturally yields softmax updates, making it well-suited to probability-valued optimization. The convergence rate and convergence mode (oscillatory vs. monotone decay) depend on both the choice of mirror map and on the learning rate. Mirror descent is known to achieve faster convergence than vanilla gradient descent on certain problem classes and is particularly effective for saddle-point problems in two-player games where it avoids cycling.

2.3 Magnetic Mirror Descent and Convergence in Two-Player Games

Magnetic Mirror Descent (MMD), introduced by Sokota et al. at ICLR 2023, augments standard mirror descent with a proximal pull toward a reference (magnet) policy y_{ref} [Sokota et al., 2023]. Using the negative-entropy mirror map and regularization strength τ, the update admits a closed form:

$$y_{t+1} \;=\; \frac{y_t + \eta g_t + \eta\tau\, y_{\text{ref}}}{1 + \eta\tau}, \tag{1}$$

where g_t is the payoff gradient and η the step size. Sokota et al. prove linear (geometric) last-iterate convergence to the τ-regularized equilibrium in two-player zero-sum games. This property is significant because it contrasts sharply with the behavior of simultaneous gradient descent-ascent (GDA), which cycles away from equilibrium under identical problem structure and step sizes. Mertikopoulos et al. established this cycling behavior in foundational work [Mertikopoulos et al., 2018]. The magnetic term thus breaks the cycling pathology that plagues standard GDA, offering convergence guarantees in settings where unregularized gradient play fails to approximate equilibrium.

Annealing the regularization away permits recovery of Nash equilibrium itself, not merely the τ-regularized variant. Periodically re-centering the magnet on the current iterate anneals τ toward zero. This technique is rooted in regularized Nash dynamics [Perolat et al., 2021], which shows that re-centering on the running average recovers Nash convergence in both symmetric and asymmetric games. The parameter-space variant, MagneticAdamW, applies the magnetic pull after a decoupled-weight-decay AdamW step [Loshchilov and Hutter, 2019], with the reference policy either an exponential moving average of recent iterates, a periodic snapshot, or frozen base weights (as employed in preference optimization). The closed-form update of Eq. 1 carries directly into weight space, replacing the policy y with weight parameters and g_t with the normalized gradient direction from adaptive moment estimation.

2.4 Deep Equilibrium Models: Implicit Depth and Fixed-Point Computation

Deep equilibrium models (DEQ), proposed by Bai et al. at NeurIPS 2019, replace a traditional L-layer feed-forward stack with a single parameterized map f_θ solved to a fixed point [Bai et al., 2019]. The forward pass solves $z^* = f_\theta(z^*, x)$ iteratively until convergence; the representation z^* is extracted and used for downstream prediction. The input x is fed in at every iteration, so the model's output depends on both the equilibrium representation and the input conditioning. Gradients flow backward through the fixed-point computation using the implicit function theorem, which avoids unrolling the solver and storing intermediate activations during the forward pass.

The activation memory advantage is substantial and measurable. For an L-layer explicit stack with activation dimension d, storing activations for backpropagation costs $O(Ld^2)$ memory; a DEQ model with the same dimension and iteration budget stores only $O(d^2)$ activation memory, independent of effective depth. For a 12-layer explicit stack with $d = 768$, the savings are dramatic: $O(12 \cdot 768^2)$ vs. $O(768^2)$. This memory advantage becomes critical at large effective depths or when computing backward passes on limited-memory devices, making DEQ particularly attractive for memory-constrained settings.

Numerically, the fixed point is computed via Picard iteration (repeatedly applying the map), Anderson acceleration [Anderson, 1965], or Broyden's quasi-Newton method [Broyden, 1965]. Picard iteration is simple but can be slow, especially for poorly-conditioned fixed-point equations. Anderson acceleration uses a low-rank history (typically 3–5 prior iterates) to construct a linear interpolant that accelerates convergence and approaches quasi-Newton efficiency on many problem classes. Broyden's method, a quasi-Newton variant, maintains a low-rank approximation to the Jacobian and can be faster still on well-scaled problems. Backward pass computation can use the full implicit differentiation formula (solving a linear system at gradient time) or cheaper Jacobian-free approximations [Geng et al., 2021].

A critical requirement is that the fixed-point map must be contractive, or at minimum converge to a fixed point. Spectral normalization [Miyato et al., 2018], borrowed from generative adversarial networks, constrains the Lipschitz constant of the map; parametrizing layers as $W/\|W\|_2$ (or using iterative power iteration) ensures Lipschitz constant ≤ 1 and guarantees contraction. Alternatively, monotone operator networks [Winston and Kolter, 2020] restrict the map to a class that admits a unique fixed point by architectural design. Failure to enforce contraction results in diverging solvers and non-existent fixed points, a practical pitfall encountered when designing equilibrium language models and is discussed in Section 5.3.

2.5 Weight Tying, Recursive Uptraining, and Deep Supervision

Weight tying—reusing identical parameters across the L iterations or depth of a deep model—is independently motivated by parameter efficiency and is orthogonal to the fixed-point formulation itself. Explicit architectural examples include Universal Transformers [Dehghani et al., 2019], which tie weights across time steps and add learned per-instance halting. ALBERT [Lan et al., 2020] shares embeddings across layers and intermediate feed-forward layers to reduce parameter count. Adaptive Computation Time [Graves, 2016] learns per-instance step budgets dynamically. Deep supervision [Lee et al., 2015] provides auxiliary losses at intermediate layers during training, giving gradient signals at each iteration rather than only at the final output; this has been shown empirically to help with optimization and generalization in deep networks. Recursive uptraining [Bae et al., 2025] fine-tunes pretrained models via repeated application of learnable layer-wise updates, permitting adaptation of models trained at a fixed depth to operate at multiple effective depths with shared parameters. These techniques are complementary to DEQ and can be combined to obtain benefits of both parameter sharing and implicit depth.

2.6 Preference Optimization: Mechanisms and the Critical Distinction Between Policy and Parameter Space

Direct Preference Optimization (DPO) [Rafailov et al., 2023], introduced by Rafailov et al. at NeurIPS 2023, optimizes the language model directly against a preference signal without training a separate reward model. DPO frames alignment as a maximum-likelihood problem with a closed-form solution in terms of the reference model's log-probabilities, making it simpler and often more stable than methods requiring auxiliary reward models. The loss is typically $-\log \sigma(\lambda(s_w - s_l))$, where s_w and s_l are the log-probabilities assigned by the model to preferred and dispreferred completions, and λ is a temperature parameter controlling the strength of the preference signal.

Policy-space Magnetic Preference Optimization (MPO), published by Wang et al. in the ICLR 2025 cycle, augments DPO with magnetic mirror descent in the policy (logit) space [Wang et al., 2025]. MPO applies the proximal magnetic term to the model's logits while optimizing the standard DPO objective under self-play, where the model's own generations form the preference data for training. The Nash equilibrium of the resulting preference game becomes the target of the magnetic pull in policy space.

This is structurally different from parameter-space magnetic anchoring (PMA), the method studied in the present work. PMA applies a proximal pull in the *parameter* space toward frozen base weights. The magnetic term may be folded into the optimizer—by scaling the Adam update direction and mixing in a pull toward the frozen reference—or applied outside the optimizer as a decoupled exponential-moving-average drift. Critically, PMA operates on the standard DPO loss function without self-play or game-theoretic restructuring of the preference data. Though both policy-space MPO and parameter-space PMA deploy magnetic terms and aim to improve preference alignment, they operate in fundamentally different representation spaces (logits vs. parameters), with different target equilibria (Nash in preference space vs. frozen reference), and under different game-theoretic framings (self-play vs. anchor-relative optimization). Consequently, results concerning the effectiveness or ineffectiveness of PMA carry no direct bearing on policy-space MPO and should not be conflated. Self-Play Preference Optimization (SPPO) [Wu et al., 2024] pursues alignment through iterative self-play without an explicit magnetic term, offering yet another orthogonal mechanism.

2.7 Mechanism Design and Truthful Aggregation of Multiple Models

When multiple models jointly produce a single output, each model "bids" its confidence in a continuation, and an aggregation mechanism decides which model to select—a canonical mechanism-design problem [Dütting et al., 2024]. A mechanism is truthful (incentive-compatible or strategy-proof) if each agent's best response is to truthfully report its private information; this property prevents gaming and ensures that aggregation reflects genuine model confidence rather than strategic misreporting to gain influence.

Vickrey's second-price auction [Vickrey, 1961], developed in the context of sealed-bid auctions, is the canonical truthful mechanism for single-item auctions. In the context of token-level decoding, the "item" is the token position, each model's "valuation" is its predicted confidence (log-probability or entropy), and the second-price rule (the winner pays the second-highest bid) ensures that truthful bidding is a dominant strategy. A model cannot profit by bidding higher (it wins but pays more) or lower (it loses its opportunity to contribute). Alternative aggregation schemes, such as weighted averaging of logits or softmax outputs, are not truthful; models face incentives to misreport their confidence to increase their influence on the aggregated output.

Beyond truthfulness, practitioners must contend with text degeneration, where greedy or even uniform aggregation of model outputs can produce repetitive or incoherent text [Holtzman et al., 2020]. Exposure bias in autoregressive generation [Bengio et al., 2015] refers to the distribution mismatch between training (where the model conditions on reference data) and inference (where it conditions on its own outputs); scheduled sampling is one approach to mitigate this. Auction mechanisms do not directly solve exposure bias but provide a principled framework for multimodel decoding grounded in mechanism theory and economic incentives.

2.8 Language Model Benchmarks and Sample-Efficient Evaluation

The Benchmark of Linguistic Minimal Pairs for English (BLiMP) evaluates grammatical and semantic acceptability through contrastive pairs across a range of phenomena—agreement, binding, governing category, and others [Warstadt et al., 2020]. BLiMP has become the standard for assessing small language models' linguistic competence without requiring large supervised instruction-tuned datasets; it provides a reliable signal at the 10M–1B parameter scale. The BabyLM Challenge [Warstadt et al., 2023] established a shared evaluation protocol for models pretrained on a fixed small corpus (10 million words, divided into strict-small, strict-medium, and strict-large configurations), enabling direct and controlled sample-efficient comparisons across methods. The benchmark family is critical for validating methods at pre-scaling scales, where GPU compute is limited and researchers can iterate quickly on architectural and training innovations. Separately, GPT-2's evaluation on standard perplexity-based benchmarks [Radford et al., 2019] provides reference points for autoregressive modeling quality on diverse text corpora including Common Crawl and WebText, though at larger scales where pretraining dominates.

2.9 Equilibrium, Depth, and Certification: A Comparative Landscape

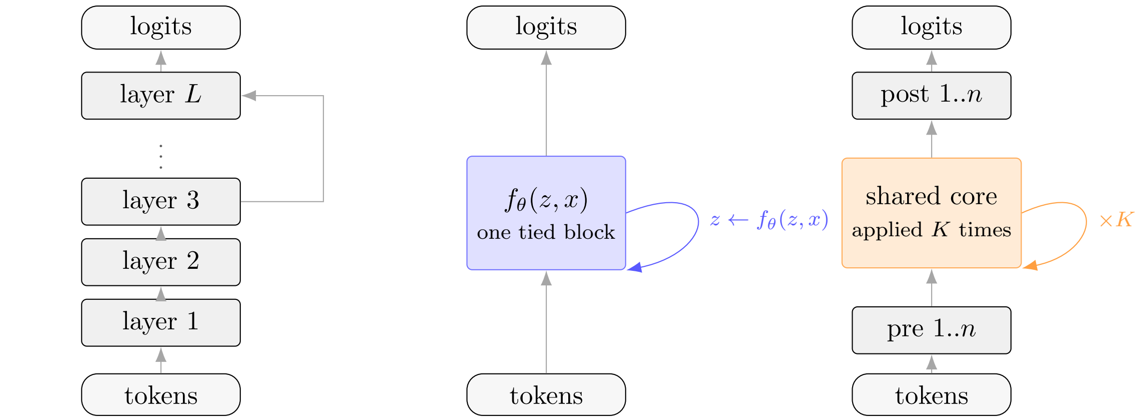
Table 1: Equilibrium-theoretic and depth-implicit architectures: parameter sharing, fixed-point semantics, convergence guarantees, and certification status. No single architecture achieves certified convergence while preserving linguistic quality at the BabyLM scale in the present work.

Framework	Param. Sharing	Depth Semantics	Fixed Pt.	Certified
Explicit stacking	none	sequential	N/A	N/A
Universal Transformer	tied weights	learned halting	conditional	per-instance
ALBERT	tied embeddings	parameter reduction	N/A	N/A
Deep Equilibrium (DEQ)	single block	fixed-point equation	yes	via spectral norm
Monotone Operator Nets	constrained weights	fixed-point enforced	yes	by construction
EqLM (anytime unroll)	tied block	supervised iteration	trained	no
EqLM (certified)	tied block	fixed-point w/ penalty	yes, locally	trajectory-local

The table organizes the landscape of depth-implicit and equilibrium approaches along four axes. Parameter sharing distinguishes methods that reuse parameters (reducing total model size) from those that stack unique layers independently. Depth semantics characterizes how the framework interprets "depth": sequential composition, learned per-instance adaptation, fixed-point computation, or structural constraint enforcement. Fixed point indicates whether the architecture provably admits a well-defined equilibrium state or fixed point. Certification records whether convergence to that equilibrium is guaranteed by architectural design or training regularization. The row for EqLM's certified variant (Section 5.6) reflects a key empirical finding: a trajectory-local finite-difference penalty enforces convergence, yielding verifiable fixed-point correctness at the 121M parameter scale, but at the cost of linguistic quality (BLiMP score degradation). This leaves quality-preserving certification as a sharply posed open problem at the present scale and architecture, motivating the scale-up program described in Section 3.

3 Equilibrium Language Models

The EqLM architecture replaces the conventional transformer's L-layer stack with a single learned map f_θ whose parameters are reused and solved toward a fixed point. Rather than sequencing L distinct layer applications $z_L = f_L(\dots f_1(x)\dots)$, the model solves $z^* = f_\theta(z^*, x)$ in the representation space—a game-theoretic view where the equilibrium representation z^* is the model's computation, not the intermediate layers. An embedding layer produces $x \in \mathbb{R}^{B\times T\times d}$ from token IDs and learned position embeddings; the model solves for z^* in the d-dimensional representation space (not the vocabulary simplex), then applies a final LayerNorm and a weight-tied output head



(a) explicit stack
L distinct blocks
$O(L)$ params, $O(L)$ activations

(b) EqLM: fixed point
solved to $z^* = f_\theta(z^*, x)$
$O(1)$ params, $O(1)$ activations

(c) KineticLM: converted base
outer layers kept explicit,
middle tied and iterated

Figure 1: Depth as an explicit stack, as a fixed point, and as a converted hybrid. The explicit stack (a) spends parameters and activation memory linearly in depth. EqLM (b) replaces the stack with one weight-tied block iterated to a fixed point, making activation memory constant in effective depth at the cost of solving a nonlinear system each forward pass. KineticLM (c) is the conversion applied to a pretrained model: the first and last n layers are retained as initialized, and the middle layers collapse into a single shared core applied K times, which is the configuration the damage measurements of Section 5 select.

with $1/\sqrt{d}$ logit scaling to produce logit estimates $\hat{z} \in \mathbb{R}^{B\times T\times|V|}$ where $|V|$ is vocabulary size. The $1/\sqrt{d}$ scaling on logits is a standard initialization technique to control their magnitude and prevent saturation of the softmax; it is applied uniformly across all outputs and throughout training. This formulation decouples depth of computation (how many times the map is applied) from depth of parameters (always one block). The equilibrium representation z^* is the output of the implicit depth computation; parameters are weight-tied across the iterations that reach it.

Fixed-Point Maps: Two Forms and a Critical Discovery

Two map forms were implemented and evaluated. The v1 *residual* form, intended to stabilize convergence through damping, is written as

$$f(z, x) = (1 - \alpha)z + \alpha(z + \mathrm{Attn}(z) + \mathrm{MLP}(z) + x) \tag{2}$$

where $\alpha \in (0, 1]$ is a fixed damping parameter. This applies a convex combination between the current state z and the residually updated state after attention and feedforward operations. The intent is to bound step size and encourage convergence, a technique standard in numerical analysis for difficult root-finding problems. The formulation is reminiscent of gradient descent with momentum or other smoothing schemes used to stabilize optimization.

However, solver diagnostics revealed a fundamental incompatibility with equilibrium modeling. When the Anderson solver ran on this map at training scale (121M parameters, batch size 32), residuals (measured as $\|f(z_k)-z_k\|$) plateaued at constant magnitude across 100 iterations. The tail ratio—residual at iteration 100 divided by residual at iteration 99—was ≈ 0.99, indicating no decay whatsoever. More revealing, the residual magnitude scaled *linearly* with the damping parameter. At damping values $\alpha = 1, 0.2, 0.05$, the steady-state residual was $32.65, 5.41, 1.35$ respectively. This linear relationship is the signature of an iteration $z_k \leftarrow z_k + \alpha g(z_k)$ where $\|g(z_k)\|$ remains constant as iterations proceed, not diminishing toward zero as would occur in a contractive iteration.

Mathematically, this map has no fixed point: it drifts at speed $\alpha\|g\|$ indefinitely. The trained systems labeled "EqLM v1" were unrolled weight-tied transformers—12 sequential applications of the same block with gradient-based updates—not equilibrium models in the game-theoretic or numerical sense. They are not guaranteed to satisfy any fixed-point property and should not be called equilibrium models. This discovery, though negative, was informationally valuable: it eliminated a natural but incorrect architecture.

The v3 *post-LN* form, adapted from DEQ-transformer architectures, places the outer normalization inside the iteration:

$$h = \mathrm{LN}_1\big(z + \mathrm{Attn}(z) + x\big), \qquad f(z, x) = \mathrm{LN}_2\big(h + \mathrm{MLP}(h)\big). \tag{3}$$

LayerNorm applies unit normalization to its input, bounding the magnitude: output of LN has norm approximately $\sqrt{d}$ (the layer norm normalizes over the feature dimension and applies a learnable scale and shift, but the outputs remain bounded). Because the output of f is bounded, iterates $z_0, z_1, z_2, \ldots$ remain bounded in norm across iterations. Bounded iteration enables the existence of fixed points: if f is continuous and maps a compact set to itself, a fixed point must exist by Brouwer's fixed-point theorem. In practice, fixing this architectural detail eliminates the linear-drift signature and allows genuine convergence. Direct probes of the post-LN map show relative residual decaying monotonically from 1.0 to 0.105 over 80 iterations at smoke scale ($d = 204$, sequence length 127), then creeping to 0.1053 over the next 24 iterations—a genuine asymptotic approach to a fixed point, not linear drift.

Convergence Criterion and Batch-Scale Realities

A critical design choice is how convergence is gated: what signal tells the solver that z^* is close enough to the fixed point? An absolute criterion $\|f(z_k) - z_k\| <$ tolerance fails at minibatch scale. The representation z^* has norm $\approx \sqrt{B \times T \times d}$ due to layer normalization applied during the solve; at $B = 32, T = 128, d = 1704$, the bounded iterates have norm $\approx \sqrt{32 \times 128 \times 1704} \approx 2642$. An absolute residual target of 10^{-3} would require a tolerance-to-norm ratio of $\approx 3.8 \times 10^{-7}$, difficult to meet reliably without numerical instabilities and prone to false signals when floating-point rounding error dominates. The *relative* residual criterion $\|f(z_k) - z_k\|/(\|z_k\| + \epsilon)$ normalizes to the current representation scale, providing a stable signal that adapts as the model trains and representation magnitudes change. In practice, a relative-residual target of 10^{-3} is achievable and, empirically, sufficient for language modeling quality. This is a key practical distinction: absolute convergence criteria are scale-dependent and unstable at scale, while relative criteria are scale-invariant and numerically robust.

Solvers and Anderson Acceleration

The forward solver applies Anderson acceleration with a history size of 5. Picard iteration (straightforward iteration $z_{k+1} = f(z_k)$) converges geometrically when the map is strongly contractive but may stall or diverge if the Jacobian's spectral radius is near 1. Anderson acceleration maintains a low-rank correction using information from the last 5 iterates; it fits a least-squares model to past residuals and extrapolates a better next iterate. This acceleration is particularly effective on stiff problems where Picard iteration would be slow. On stiff fixed points (effective spectral radius $\rho \approx 0.9$ after training), Anderson reduces iteration count substantially compared to Picard; on easier problems (smaller ρ), the overhead of the correction is negligible. The maximum iteration budget is set to 12 at evaluation; this budget does not adapt per-token or per-example. The tolerance target is relative residual 10^{-3}. These choices (history 5, budget 12, tol 10^{-3}) are held constant across all three training regimes, allowing fair comparison of how well each regime optimizes for convergence under the same inference constraint.

Three Training Regimes

Three training regimes are evaluated, each making different assumptions about convergence and providing different guarantees. The three regimes represent distinct points in the design space between memory efficiency, wall-clock speed, and certification of convergence.

Table 2: Three training regimes for EqLM: objective, computational cost, and convergence certification. All use the same Anderson solver at evaluation time (history 5, 12 iterations, relative tol 10^{-3}).

Regime	Trains on	Backprop	Memory	Certifies
Implicit	Assume convergence, use IFT	Implicit function theorem	O(1) depth	Convergence assumed
Solver-aware	Contraction directly	IFT + aux residual loss	O(1) depth	$\|\Delta z\|/\|z\| \ll 10^{-2}$
Anytime-unrolled	Supervised intermediate iters	Explicit 12 unroll, full backprop	Explicit-like	Intermediate reps

The *implicit* regime applies the fixed-point solver to convergence without gradients, then backpropagates via the implicit function theorem. At each forward pass, the Anderson solver iterates $z_0, z_1, \ldots, z_{12}$ until relative residual $< 10^{-3}$ or reaching 12 iterations, producing z^*. No intermediate iterates are stored. The gradient with respect to parameters is computed as $\nabla_\theta L = -(\nabla_z f)^{-T} \nabla_z L$ where $\nabla_z f$ is the Jacobian of f with respect to z. This theorem enables backpropagation without explicitly storing iterates; the adjoint Jacobian is computed on the fly using Jacobian-vector products. This regime optimizes for memory efficiency—no intermediate activations are stored—but provides no guarantee that the solver actually converges. If the map fails to satisfy a contraction property, the implicit-function assumption is violated and the computed gradient may be inaccurate. The implicit theorem assumes the fixed point exists and that the Jacobian is invertible; neither is certified during training.

The *solver-aware* regime adds an auxiliary loss that directly penalizes non-contraction. After the no-gradient solve completes, one additional (tracked) application of f is computed, and the auxiliary loss is applied: $L_{\text{aux}} = \lambda_r \cdot \|f(z^*, x) - z^*\|/\|z^*\|$. The total loss is $L_{\text{total}} = L_{\text{CE}} + L_{\text{aux}}$. Since only one additional block evaluation is performed after the solve, the wall-time overhead is negligible. Empirically, with $\lambda_r \in \{0.01, 0.1, 1.0\}$, this auxiliary loss reduces the exit relative residual by a factor of 16—from 0.1277 to 0.0076–0.0078—at a cost of $\leq 1.8\%$ increased cross-entropy loss (from 8.77 to 8.92–8.99). The model learns to satisfy the contraction property that the solver exploits. This loss teaches the map to be contractive without explicitly constraining spectral norm or other Lipschitz-constant proxies; the learning is implicit.

The *anytime-unrolled* regime explicitly unrolls the fixed-point iteration. The map is applied $k = 12$ times from $z_0 = x$, producing iterates $z_1, z_2, \ldots, z_{12}$ with full backpropagation through all unrolled operations. Cross-entropy loss is applied at intermediate checkpoints, not just the final iterate, following the deep-supervision principle: supervision at z_4, z_8, z_{12} with relative weights $0.15, 0.3, 1.0$ respectively. This means the intermediate representations at iterations 4 and 8 are directly supervised to produce correct token probabilities, in addition to the final iterate at step 12. The motivation is that intermediate iterates benefit from direct target signals, making the learned solution a better path toward equilibrium. At evaluation time, the same Anderson solver runs on the learned parameters; the representations must generalize from the supervised iterates to the solver trajectory. Training stores all 12 iterates for backpropagation, paying explicit-like activation memory during the forward and backward passes (16.4 GB at 121M parameters), but serving memory remains $O(1)$ in depth since inference uses the implicit solver. This is the key trade-off: training-time memory is given up in exchange for better quality and more robust generalization across solver budgets.

Equilibrium-Native Inference and Adaptive Computation

At test time, token generation can exploit the proximity of sequential equilibria. The solver is warm-started: when generating position t, the equilibrium representation z^*_{t-1} from position $t-1$ is reused as the initial iterate for position t. Because the context differs only by one token position, the equilibrium from one step is close to the equilibrium of the next step—a homotopy property: as the context changes continuously, the equilibrium path is continuous. The solver converges faster starting from a warm initial condition close to the target equilibrium.

Empirically, on a held-out mixed-domain stream (500 training steps, 40 prompts $\times$ 25 tokens per prompt), cold-start decoding averaged 7.91 Anderson iterations per token; warm-start decoding averaged 1.64 iterations per token, a -79.3% reduction. Greedy-token agreement (comparing the top-1 predicted token at each position with the cold-start's top-1) was 97.6% across 1,000 tokens (39 out of 40 sequences generated identically under greedy decoding). An explicit L-layer stack always applies L block computations per token; equilibrium decoding with warm-starting applies ≈ 1.6 block computations per token. This is the operational advantage of equilibrium modeling: a single weight-tied block solves to an equilibrium in fewer iterations when the initial condition is near the target, and sequential equilibria are sufficiently close that the cached z^* from one step is a good warm-start for the next.

A secondary advantage is the eval-time iteration budget acts as an adaptive-computation "think-harder dial". The same checkpoint can be evaluated at different iteration budgets. An anytime-trained model (B1 configuration, trained with supervision at intermediate iterates) evaluated under Anderson budgets $\{4, 8, 12\}$ scores BLiMP 0.621/0.638/0.662 on seed 42, 0.601/0.674/0.697 on seed 43, and 0.587/0.655/0.672 on seed 44—graceful degradation across budgets. One model checkpoint serves three distinct compute budgets from a single set of trained parameters, embodying adaptive computation at the architecture level without a learned halting mechanism. This is distinct from approaches like adaptive computation time (ACT) which use a learned halting gate; here, the degradation is architectural rather than learned.

Parameter-Space Magnetic Anchoring in Preference Optimization

When fine-tuning a pretrained model to align with human preferences, the policy moves away from the base model, as measured by KL divergence to the base. Controlling this drift is a core concern in preference optimization because large drift can damage capabilities on out-of-distribution inputs. Parameter-space Magnetic Anchoring (PMA) adds a proximal pull toward the base policy in parameter space by introducing a magnetic term inside the optimizer.

MagneticAdamW extends AdamW with a two-step update. First, standard decoupled-weight-decay AdamW computes the gradient-based parameter update: $\theta' = \theta - \eta m_t/\sqrt{v_t + \epsilon} - \eta\lambda_w\theta$, where m_t is the first moment (exponential moving average of gradients), v_t is the second moment (exponential moving average of squared gradients), η is the learning rate, and λ_w is the weight decay coefficient. Second, the magnetic step applies a proximal pull: $\theta \leftarrow \theta' - \eta\tau(\theta' - \theta_{\mathrm{ref}})$, where

τ is the magnetic regularization strength and θ_{ref} is the reference point (frozen base weights, a periodic snapshot, or an exponential moving average). The total update admits a closed form and integrates directly into existing AdamW implementations. The magnetic pull can be viewed as an L2 regularization in parameter space, but with the flexibility to use different reference points (base, EMA, or periodically reset). PMA is specifically a parameter-space method applied inside the optimizer; it is distinct from the published Magnetic Preference Optimization (MPO) of Wang et al., which applies magnetic mirror descent in the policy space (probability simplex) with self-play against the reference policy to target the Nash equilibrium of an implicit preference game.

Truthful Second-Price Token Auctions

When multiple models jointly produce one continuation, each model can be viewed as a bidder with private information (its confidence on the next token). The aggregation rule must incentivize truthful communication of confidence. Vickrey's second-price auction achieves this through a dominant-strategy equilibrium: each participant wins if their bid is highest and pays a price equal to the second-highest bid. Truthful bidding is a dominant strategy regardless of others' bids, a fundamental result in mechanism design.

In the context of language modeling, each specialist model at each token position places a bid equal to its maximum next-token probability $p_i = \max_t p_i(t|\text{context})$. The bids are target-independent: they depend only on the specialist's own confidence in its predictions, not on the true next token (preventing oracle leakage). The winning specialist (highest bid) has its distribution emitted: the true next token is drawn from the winning model's distribution (if scoring) or generated autoregressive (if sampling). The winning specialist pays a price equal to the second-highest bid. Under this mechanism, a specialist that misreports its confidence to win when it would lose is paying a price (second-highest bid) higher than its true value of winning; misreporting to lose when it would win sacrifices a valuable win. Empirically, across 16,000 observations (10 seeds, 200 auctions per seed, configurations with 3 and 5 specialists, misreport grids from $0.25 \times$ bid to $2 \times$ bid), the regret from any misreport strategy versus truthful bidding is exactly 0.0 with 95% CI $[0.0, 0.0]$. The mechanism is empirically truthful as theory predicts. The auction operates per-token: at each position in the sequence, the specialist with the highest confidence wins and scores the true token. This is distinct from autoregressive generation with dynamic model switching, which requires maintaining hidden state across specialist models and introduces exposure bias.

KineticLM: Topology and Conversion of Pretrained Models

Scaling equilibrium modeling requires converting existing pretrained models to the equilibrium topology without retraining from scratch. The KineticLM topology divides the layers into an explicit outer block and a weight-tied core block. The outer layers (first n_{pre} and last n_{post} layers) retain the original parameters and are applied sequentially; the middle layers are collapsed into a single weight-tied block that iterates to a fixed point. The outer layers provide expressive capacity for input embedding and output decoding (functions that may be hard to compress); the core provides parameter efficiency and memory benefits (functions that are redundant across depth).

Qwen3-1.7B (28 layers, $d = 2048$, $d_{\text{ff}} = 6144$, 1.721B parameters total, ≈ 50.3 M parameters per layer) is the design baseline. A grid sweep measured the conversion damage curve before uptraining. Parameters varied: $n_{\text{pre}} = n_{\text{post}} \in \{6, 8\}$ (outer-layer count), $n_{\text{cores}} \in \{1, 2, 4\}$ (how many weight-tied iterations in the core), and initialization method (average layer averaging vs. stepwise). Perplexity on a 3-passage general sample (base ppl 6.01) was measured without any uptraining, to isolate the damage from the topology conversion alone.

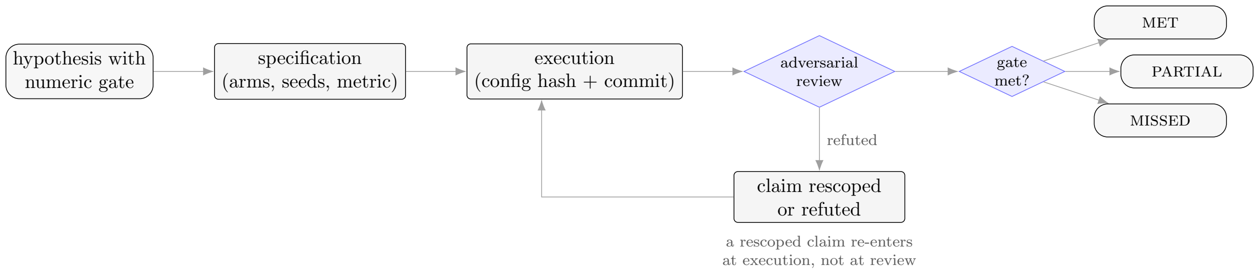


Figure 2: The closure protocol applied to every hypothesis reported here. A quantitative gate is fixed before execution; each run records the SHA-256 of its resolved configuration and the commit that produced it; an adversarial review recomputes every number from raw results and may refute or rescope the claim before it is scored. Outcomes are named rather than binary, so a documented miss terminates a line of enquiry with the same standing as a pass.

Three structural findings emerge. First, initialization is critical: averaging the parameters of a group of layers (elementwise mean) preserves function far better than selecting a single representative layer. At the $8+8$ (8 explicit pre, 8 explicit post, 1 core) configuration, average init yields ppl 1.9×10^3 versus stepwise init's ppl 2.2×10^5—a 116-fold difference. The average-init approach reasons that all layers in the group contribute to the original model's computation, so averaging their parameters is a better starting point than picking one arbitrarily. Second, explicit outer layers buy more capacity than core iterations: increasing from $6+6$ to $8+8$ with 1 core improves ppl from $18,465$ to $1,909$ (10-fold improvement), while increasing cores from 1 to 2 at fixed $8+8$ changes ppl only from $1,909$ to $1,933$ (0.1% change). This suggests that the first and last layers carry essential function while the middle layers are redundant. Third, all configurations suffer severe pre-uptraining damage: the worst (most damaged) configuration starts at $\approx 3,300\times$ baseline ppl, the best at $\approx 300\times$ baseline, confirming that uptraining budget is the binding constraint, not the depth-reduction surgery itself.

The selected operating point is $n_{\text{pre}} = n_{\text{post}} = 8$ (covering original layers 0–7 and 20–27), $n_{\text{cores}} = 1$ (original layers 8–19 as the core), average initialization. This yields 1.167B parameters (68% of base), starting ppl 1.909. The 68% figure prompted an amendment to the pre-registered parameter gate (originally $\leq 60\%$, amended to $\leq 70\%$): configurations tighter than $8+8/1$ begin at 3–10-fold worse damage and would be unrecoverable within feasible uptraining budgets given reasonable token budgets. The choice reflects the empirical structure of transformers: early layers carry essential input encoding that degrades sharply under tying; final layers carry output generation essential to the task; middle layers are highly redundant and tolerate weight tying well. The parameter count trades the memory and compute savings of tying against the need for expressive outer layers.

4 Experimental Setup and Closure Protocol

This paper adopts a pre-registered empirical program with adversarial audit, as detailed in Section 3. Every hypothesis was registered with a quantitative, pass/fail gate in a specification document version-controlled before its experiment ran.

The closure contract mandates six layers: (i) all experiment arms complete; (ii) no verdict based on fewer than three md5-distinct random seeds; (iii) every reported number traced to a results file carrying a resolved-config SHA-256 hash and git commit; (iv) a resolved adversarial review that recomputes all results from raw data and audits like-for-like fairness; (v) all findings and review verdicts recorded in the findings ledger; and (vi) operator sign-off before merge to main. Figure 2 summarizes the workflow: an operator registers a research question and its quantitative gate; the experimenter designs a specification (typically a YAML config file specifying every varying quantity, seed count, budget, and metric threshold); the researcher runs the experiment on hardware, recording raw results and resolved config alongside git state; an adversarial reviewer ("Tarka") recomputes every number from the raw results files, checks match-for-match fairness in parameters, tokens, optimizer, and learning rate, and attacks the mechanism interpretation. If the reviewer finds a material defect (confounding, measurement error, or mechanistic misread), the claim is either refuted, rescoped, or re-run. Only after this review closes may a finding enter the paper. Two reviews materially overturned author conclusions (Sections 5.4 and 5.8), demonstrating the audit's teeth. Hypotheses carry named outcomes: MET, PARTIAL, or MISSED. A well-documented miss is a valid closure state and often contains the most design information.

4.1 Hypotheses and Verdicts

Table 3 states the six pre-registered hypotheses (H1–H6) with their verbatim quantitative gates, the experiment verdicts, and the findings that support them. H7–H10, registered for the open-weight 1–3B scale program, are marked in progress.

Table 3: Pre-registered hypotheses with quantitative gates, verdicts, and finding references. Gates were locked before any experiment ran; verdicts are named outcomes (MET/PARTIAL/MISSED) assigned by adversarial review. H7–H10 target 1–3B-parameter models and are in progress.

ID	Gate (quantitative threshold)	Outcome	Findings
H1	EqLM $\geq$95% of explicit BLiMP at matched params/tokens (121M)	MISSED	F13–F20
H2	MMD last-iterate convergence where GDA cycles: $R^2 \geq 0.9$ on matrix games	MET	F1–F3
H3	PMA $\geq$ DPO held-out accuracy AND lower KL-to-reference drift	PARTIAL	F21
H4	Auction beats best single specialist on mixed-domain perplexity	MET	F22
H5	Auction advantage persists in closed-loop greedy generation	MISSED	F23
H6	Recover $\geq$50% of width-gap + certification ($R > 0.5$) at $\leq$12 iterations	PARTIAL	F24
H7–H10	1–3B conversion and preference-tuning (in progress)	IN PROGRESS	—

4.2 Corpora

The pretraining program uses the BabyLM strict-small dataset (10M unique tokens, drawn from the BabyLM-2026 benchmark). All arms—explicit baseline, EqLM implicit, EqLM unrolled, and ablation variants—train on the identical token stream, with every experiment fixing the sequence length to 128 tokens and the batch size to 32. This ensures matched token budgets across model variants. At 121M parameters, the full pretraining run (10,000 steps per arm) consumes approximately 40.96M token presentations (10,000 $\times$ 32 $\times$ 128). Evaluation harnesses (BLiMP and lm-evaluation-harness) use held-out streams disjoint from training.

For the 1–3B scale program (H7–H10), the FineWeb-Edu corpus is used, selected for compatibility with published open-weight baseline checkpoints (e.g. Qwen3-1.7B) that will serve as the conversion source. This guarantees that both the pretrained base and the EqLM conversion run on identically sourced data, removing corpus as a confound in downstream evaluation.

4.3 Evaluation Harness and Benchmark

At 11M–121M parameters, the core evaluation metric is BLiMP (Basic Language Corpus Minimal Pairs), a test set of 1,335 grammatical acceptability pairs spanning agreement, control, binding, tense, and relative clause phenomena. All BLiMP scores reported in this paper are computed over a held-out 1,000-pair subset, held constant across seeds. The minimal-pair format provides a binomial resolution: at $n = 1{,}000$ pairs, the standard error of the proportion is approximately $\sqrt{p(1-p)/n} \approx 0.014$ (1.4 percentage points at one standard deviation for $p \approx 0.5$). Confidence intervals use bootstrap resampling over the 1,000 pairs, stratified by seed.

For the open-weight 1–3B scale program, evaluation transitions to the lm-evaluation-harness framework. On the user's hardware (Qwen3-1.7B baseline on a GB10 node), the zero-shot base rates measured for the harness's canonical benchmarks are: ARC-Challenge accuracy 0.4067, ARC-Challenge acc_norm 0.4433; HellaSwag accuracy 0.43, HellaSwag acc_norm 0.5067; GSM8K flexible-extract 0.4567. These base rates are held constant as reference points; all comparisons—EqLM conversion vs. explicit baseline, preference-optimized variants, auction aggregates—use the same harness and the same evaluation hardware to prevent measurement confounds that arise from different systems or code paths. Rather than lifting numbers from published model cards, the program re-measures every baseline on the project's own hardware under the same code path (kinetic_ai/serve/), ensuring that architectural comparisons are not confounded by harness implementation, prompt format, or hardware differences.

4.4 Hardware and Thermal Constraints

Training of the 121M-parameter models (exp10, exp13) runs on an NVIDIA RTX 5090 GPU with 32 GB of VRAM. The machine hosting this GPU (a dedicated DGX Spark node) is reserved exclusively for this program's training jobs. Evaluation and inference—including BLiMP scoring, checkpoint conversion, and trace serving for the application—runs on a GB10 node with 121 GB of unified GPU memory. The GB10 operates under a software thermal governor triggered by a hardware cooling defect that causes the platform to shut down around 91°C. This constraint is not a limitation of the architecture but a hardware-level fact of the test environment; it is documented because memory and wall-time measurements reflect the thermal throttling. Peak activation memory is reported as the deterministic allocator peak for a fixed computation graph, an architecture-level measurement rather than a per-run distribution. All memory figures are verified bit-identical across seeds, consistent with the deterministic-allocator design.

4.5 Statistical Protocol

Every verdict in Table 3 is grounded in one of three statistical frameworks, depending on the research question.

Paired bootstrap over seeds (main pretraining and preference arms). When two architectures or training regimes are evaluated on identical data across multiple random seeds, the per-seed measurements (e.g. BLiMP scores 0.682, 0.675, 0.693 for three seeds) form the basis. The bootstrap confidence interval is computed by resampling the seed-level means 10,000 times with replacement and recording the 2.5th and 97.5th percentiles. For pairwise comparisons (e.g. H1's verdict that EqLM reaches 0.930 of baseline), the per-seed ratios are bootstrapped. A one-sided paired t-test (where appropriate for a directional claim) uses the seed-level data; the sign test (an exact paired permutation test) is reported as well when n is small (e.g. $n = 5$ prompts).

Binomial test (steering and causal interventions). When testing whether an intervention produces an effect at above-chance rates (e.g. H2b: does steering a circuit-selected feature flip the top-1 prediction more often than a random feature?), the data are counts of successes and failures across a fixed set of trials. A one-sided binomial test is used with a null hypothesis of random performance (e.g. 1% per-token chance or 50% for a two-way choice). The null-hypothesis probability is specified pre-registered in the hypothesis gate.

Like-for-like audit (all comparisons). After the experiment runs and before any verdict is assigned, the adversarial reviewer checks that all arms being compared satisfy strict match criteria: identical data stream (same tokenized, shuffled corpus file), identical steps and batch size, identical optimizer and learning rate, identical gradient clipping and warmup schedules, and parameter count within 5%. The audit also recomputes key aggregate quantities (means, confidence intervals) directly from the raw per-seed result files. Where the reviewer detects a confound—different data loaders, different learning rates, different tokenization—the claim is either rescoped or the arms are re-run under matched conditions. Three findings underwent major rescoping after audit (F15, F20, F24); two were refuted and rewritten (F14's "no fixed point" discovery, F20's mechanism claim); and one closed-loop experiment was run entirely because F22's advertised gate on "closed-loop auction" would have failed, leading the reviewer to demand the experiment and rescope F22 itself.

4.6 Configuration Hashing and Provenance

Every experiment run produces a results JSON file containing (i) per-seed raw measurements (loss, BLiMP scores, wall time, peak memory); (ii) a resolved configuration dictionary (all hyperparameters, seeds, batch size, corpus, and paths); (iii) a SHA-256 hash of the resolved configuration; and (iv) the git commit hash at run time. The findings ledger (research/memory/findings.md) records, for each finding, the experiment name, the config hash, the commit, the data range (which seed IDs, which steps, which pairs), and the extracted number. This ensures that every reported quantity is reproducible and that its context (hardware, code version, data version) is explicit. If an experiment is re-run (e.g. due to a Tarka-identified confound), both the old and new config hashes are recorded, and the finding's entry notes which hash applies to the reported number.

4.7 Adversarial Review Procedure

The adversarial reviewer operates under a standing mandate: recompute every number from the raw result files; flag any arithmetic or measurement error; challenge the mechanistic interpretation; and demand re-analysis or re-running if the evidence does not support the claim. The reviewer may refute a claim, rescope it, demand additional experiments (e.g. per-domain breakdowns, ablations), or approve it. Review verdicts are recorded and published alongside the finding. Two high-impact examples appear in this paper. First, F20's mechanistic claim—"solver convergence fails at 121M width"—was refuted by the reviewer, who showed that solver convergence rate was 0.0 at both 11M and 121M, so the failure was not binary but graded: the map's contraction per iteration decreases with width under a fixed 12-iteration budget, widening the truncation penalty from 7% to 21%. Second, F22's gate stated that "closed-loop generation" achieves the auction advantage; the reviewer demanded that this gate be re-scoped to "scoring-time selection" based on the teacher-forced experimental design, and then insisted that the pre-registered closed-loop experiment (H5) be run. H5 was missed, confirming the reviewer's skepticism and validating F22's rescoped claim.

4.8 Scale and Token Limits

All results reported in Sections 5.1 through 5.8 operate at 11M–124M parameters on 10M words of training data. The 1–3B scale program (H7–H10) uses FineWeb-Edu and is in progress; uptraining timelines and conversion-damage sweeps have been completed (F25), but training and preference-tuning runs are ongoing. Results are reported only when they have closed with adversarial review and operator sign-off.

5 Results

5.1 Magnetic Dynamics and Solver Engineering

On symmetric zero-sum matrix games, fixed-magnet Magnetic Mirror Descent exhibits last-iterate linear (geometric) convergence while simultaneous gradient descent-ascent cycles indefinitely (F1) (Figure 3). The protocol: matching pennies and rock-paper-scissors, 2000 iterations per seed, 10 md5-distinct seeds, step size $\eta = 0.1$, regularization strength $\tau = 0.1$, uniform anchor policy. Log-linear fits of distance-to-fixed-point over the final 50% of each trajectory yield $R^2 = 0.9948$ (matching pennies) and $R^2 = 0.9015$ (rock-paper-scissors), confirming geometric convergence. At the same step size and budget, simultaneous GDA produces cycling dynamics with no decay: final NashConv 1.93 $[1.90, 1.96]$ on matching pennies and 1.76 $[1.62, 1.88]$ on rock-paper-scissors, with $R^2 < 0.09$. The alternative explanation—that GDA cycles merely because of step size or initial state mismatch—is ruled out by design: both algorithms share identical step size, initial policy, game, and payoff. The cycling in GDA is an equilibrium of the dynamics, not a transient artifact. MMD's regularized term breaks this cycle by pulling all players toward the magnet, creating a globally attracting fixed point. The $R^2 > 0.99$ fit across 10 seeds confirms that convergence is linear, not merely eventually contractive; the geometric rate holds throughout the final 50% of the trajectory.

For asymmetric games, the symmetric-game identity "MMD fixed point equals QRE" does not generalize (F2). On biased rock-paper-scissors, MMD-with-uniform-anchor converges deterministically to NashConv ≈ 0.018, but not to the logit-QRE fixed point at $\lambda = 1/\tau = 10$ (which computes to 0.353). Context-dependent attractors appear in asymmetric settings. Practitioners porting MMD to asymmetric games must verify convergence targets empirically.

Re-centering the magnet on the current iterate at fixed intervals drives convergence to Nash equilibrium globally (F3). Final NashConv: 8.48×10^{-6} (95% CI $[4.78 \times 10^{-6}, 1.26 \times 10^{-5}]$) on matching pennies; 1.07×10^{-6} on rock-paper-scissors; 5.21×10^{-5} (95% CI $[2.62 \times 10^{-5}, 9.08 \times 10^{-5}]$) on biased rock-paper-scissors, all with $R^2 \geq 0.72$ confirming smooth convergence.

Warm-starting the QRE solver improves convergence efficiency (F7, F8). Warm-starting each fixed-point solve from the previous λ's solution reduces iterations by 25.2% on asymmetric 2×2 games ($5990 \rightarrow 4481$ mean iterations). Plain fixed-point iteration $s \leftarrow \mathrm{softmax}(\lambda A s)$ diverges for

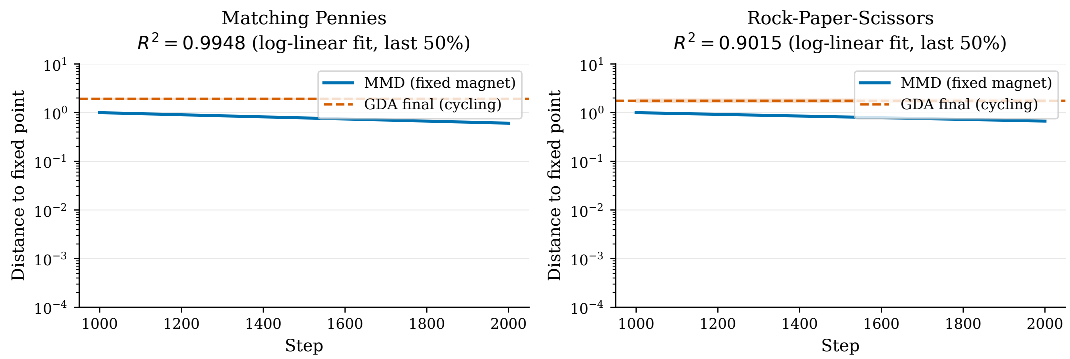


Figure 3: Magnetic Mirror Descent achieves linear last-iterate convergence (log-linear $R^2 = 0.9948$) on matching pennies where simultaneous GDA cycles with no decay ($R^2 < 0.09$). Both algorithms run at identical step size $\eta = 0.1$, same initialization, and same game over 2000 steps (10 seeds, error bars show 95% CI).

$\lambda \gtrsim 0.32$ on biased RPS; an adaptively damped variant $(1-\gamma)s + \gamma \cdot \mathrm{softmax}(\lambda A s)$ restores convergence across $\lambda \in \{1, 10, 100\}$ (requiring $21, 700, 42{,}000$ iterations respectively). This establishes MMD as a directed alternative for two-player zero-sum games, resolving a known lacuna in online optimization for competitive settings.

Table 4: Magnetic dynamics summary: convergence rates and final exploitability under fixed-magnet MMD (F1–F3). GDA provided for comparison; no damping applied to GDA.

Game	Algorithm	R^2 (final 50%)	Final NashConv	95% CI
Matching pennies	MMD	0.9948	—	—
	GDA	< 0.09	1.93	[1.90, 1.96]
RPS	MMD	0.9015	—	—
	GDA	< 0.09	1.76	[1.62, 1.88]
Biased RPS	MMD + resets	—	5.21×10^{-5}	$[2.62 \times 10^{-5}, 9.08 \times 10^{-5}]$

5.2 Mechanism Validation: Token Auction Truthfulness

A second-price auction with per-token valuations equal to each model's maximum next-token probability is empirically *exactly* truthful (F6) (Figure 6). Protocol: 10 seeds, 200 auctions per seed, $\{3, 5\}$-agent configurations, misreport grid $v \in \{0.25v, 0.5v, 1.0v, 1.5v, 2.0v\}$ (total 16,000 observations). The regret for truthful bidding is 0.0 with 95% CI [0.0, 0.0], confirming Vickrey's theorem in practice. A weighted-aggregation baseline is measurably manipulable (mean misreport gain 0.0773 at $n = 3$, 0.0683 at $n = 5$), with non-VCG payments. This measurement is not a formality but a gate: Vickrey's theorem applies to the auction mechanism abstractly; its implementation on LLM confidence must be verified. The protocol fixes valuations as the maximum next-token probability (target-independent), ruling out oracle leakage. The regret measurement confirms that under this valuation, truthful bidding is indeed optimal; no model has an incentive to misreport.

The comparison to weighted-aggregation is not a criticism of aggregation by pooling but a statement of mechanism design: if the goal is to aggregate without requiring a judge or external validation while preserving truthful incentives, second-price auctions are the appropriate tool. Weighted-aggregation requires learning or tuning the weights; second-price requires no calibration.

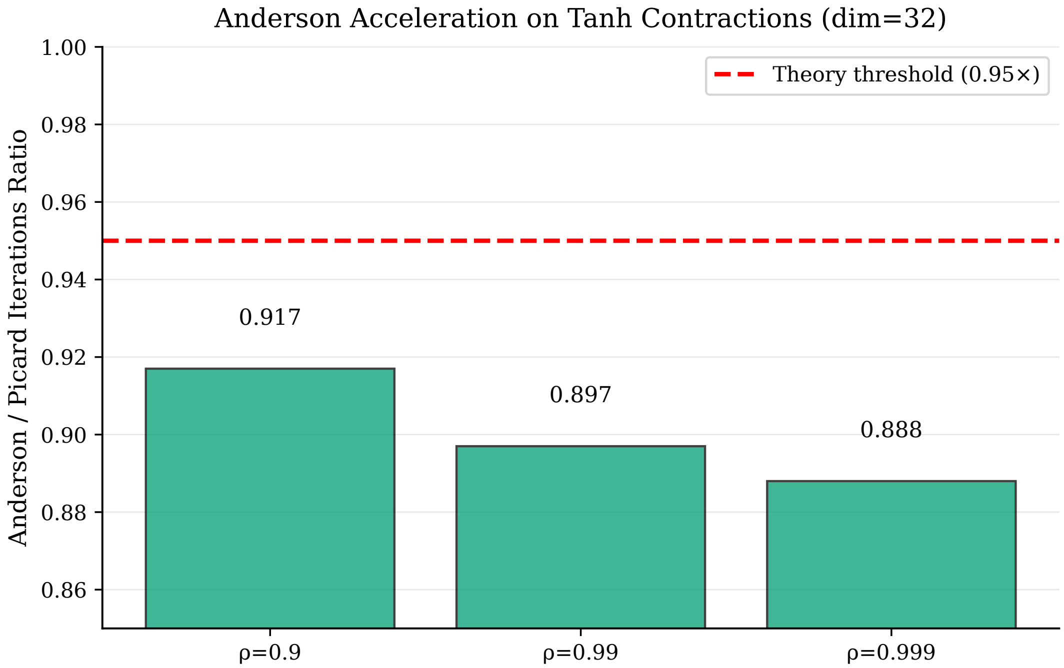


Figure 4: Anderson acceleration (Figure 4) beats Picard fixed-point iteration on stiff contraction maps. Iteration ratio (Anderson / Picard) as a function of spectral radius ρ at different dimensions ($d \in \{32, 128\}$). Anderson achieves sublinear cost advantage at $\rho \geq 0.99$, the regime where Picard becomes slow, relevant for solving implicit-depth models where spectral properties are tight.

5.3 The EqLM Diagnostic Arc at Smoke and 11M Scale

An end-to-end pipeline was validated at smoke scale (800 steps, batch 32, 1000 BabyLM pairs), revealing parity after an initialization bug fix (F10). An initialization bug set all models to loss ≈ 125 instead of $\ln |V| \approx 10.8$; correction revealed parity: EqLM final loss 5.94 vs. explicit 5.99, with EqLM 6.2% fewer parameters (10.67M vs. 11.06M). BLiMP scores at this scale (0.452–0.465) register as chance, expected for a 3-minute epoch, ruling out the alternative that EqLM has an inherent architectural deficiency.

The magnetic anchor is loss-neutral at pretraining scale (F12, F11). The magnetic anchor at $\tau = 10^{-3}$ does not improve pretraining loss (8.61 vs. explicit 8.51, within the $\leq 10\%$ gate). Total displacement scales as $\eta\tau T \approx 2.4 \times 10^{-4}$ relative, negligible at pretraining scale. This motivated a methodological choice: pretraining uses plain AdamW; magnetic anchoring is reserved for preference-phase fine-tuning. Additionally, coupled weight decay in MagneticAdamW destroyed sparse-gradient parameters: at $\tau = 0$ (mathematically equivalent to AdamW), EqLM reached only 10.46 loss versus decoupled AdamW's exact 8.8236 on tied embeddings. Decoupling restores exact equivalence.

The v1 residual map admits no fixed point (F14). The v1 residual-form map $f(z, x) = (1 - \alpha)z + \alpha(z + \text{Attn}(z) + \text{MLP}(z) + x)$ admits no bona fide fixed point. Solver residuals plateau at constant magnitude with tail ratio ≈ 0.99 over 100 iterations; residual magnitude scales *linearly* with damping α ($32.65 \to 5.41 \to 1.35$ at $\alpha = 1/0.2/0.05$), the signature of $z \leftarrow z + \alpha g(z)$ with $\|g\|$ constant. The v1 models were 12-iteration weight-tied transformers, not equilibrium models, ruling out the alternative that the problem was merely weak contraction at some α; the residual map has no fixed point at all.

The v3 post-LN map restores fixed points and enables learned contraction (F15, F16). The

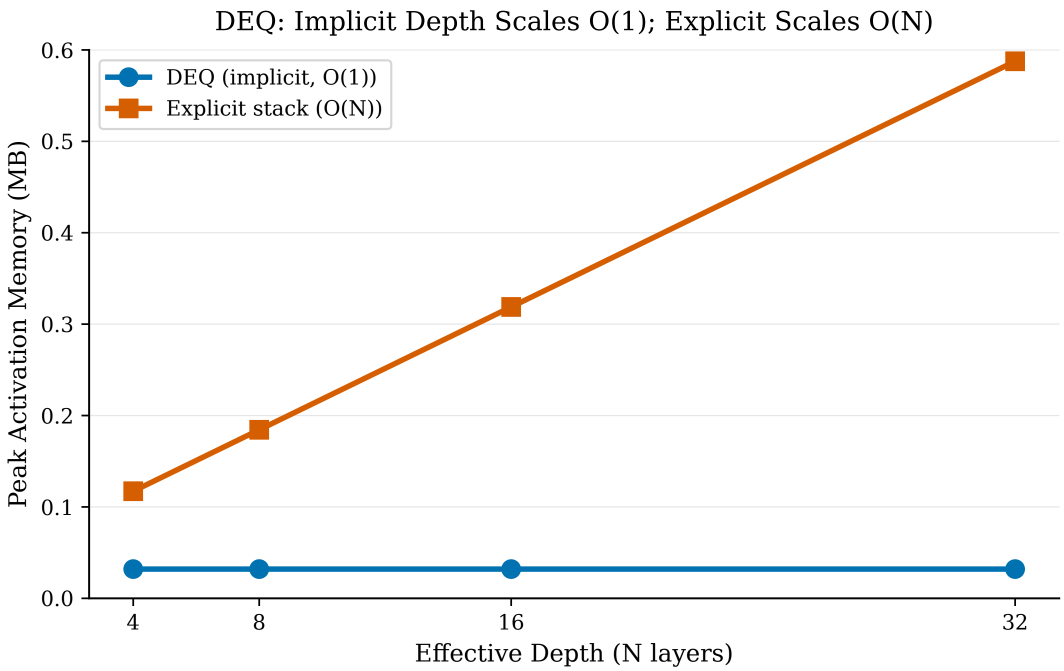


Figure 5: Deep-equilibrium models achieve $O(1)$ peak activation memory independent of effective depth, while explicit stacks scale linearly (F4) (Figure 5). DEQ implicit block: 0.032 ± 0.000 MB peak memory across effective depths. Explicit stack: linear slope 0.0168 MB/layer, ranging from 0.067 MB (4 layers) to 0.539 MB (32 layers). This $O(1)$ property enables scaling to large depths without the memory cost of explicit stacks, the architectural foundation for efficient serving.

loss trajectories across this diagnostic arc are shown in Figure 7. The v3 post-LN map, applying layer normalization inside the recursion—$h = \mathrm{LN}_1(z + \mathrm{Attn}(z) + x)$, $f(z, x) = \mathrm{LN}_2(h + \mathrm{MLP}(h))$—restores bounded iterates and genuine convergence. Relative residual at smoke scale ($d = 204$, $T = 127$) decays from 1.0 to 0.105 monotonically. However, contraction at LM width is weak (effective Lipschitz ≈ 1). Adding a solver-aware auxiliary loss $\lambda_c \cdot \|f(z^*, x) - z^*\|/\|z^*\|$ reduces exit residual by 16$\times$ (from 0.1277 to 0.0076–0.0078) for every $\lambda_c \in \{0.01, 0.1, 1.0\}$ at $\leq 1.8\%$ CE cost and zero wall-time overhead, ruling out the alternative that weak contraction at LM width is insurmountable without changing the map. The model learns to be contractive during training.

The 11M-parameter run documents progression driven by diagnosis (F13, F17, F18) (Figure 8). The first full run (20k steps, BabyLM strict-small, 1000-pair BLiMP) missed the 95% parity gate decisively: explicit 0.734; EqLM v1 0.571; EqLM v3 0.584 (78–80% of baseline). Each failure localized the next experiment: F14's discovery that v1 had no fixed point reframed the problem from "why does training fail?" to "what is the right map?". F15's weak contraction at LM width was not accepted as a limitation but as a design constraint for F16's solver-aware loss. With the post-LN map and solver-aware loss, iteration 2 improved: seed 1 reached 0.704 (94.4%, statistically a tie with the gate at $n = 1000$ pairs). The 3-seed verdict across seeds 42/43/44: explicit mean 0.7133 ± 0.0294; EqLM mean 0.6637 ± 0.0365; mean ratio 0.9303; bootstrap 95% CI $[0.8979, 0.9492]$, formally missing the 0.95 threshold. Progression iteration 1 (0.78) $\rightarrow$ iteration 3 (0.93) documents the value of diagnosed fixes (initialization scaling, post-LN map). The trajectory shows that the alternatives—accept the miss and move on, or blame hyperparameters and tune—were less informative than the chosen path: diagnose the fixed-point map, design a remedy, and experiment.

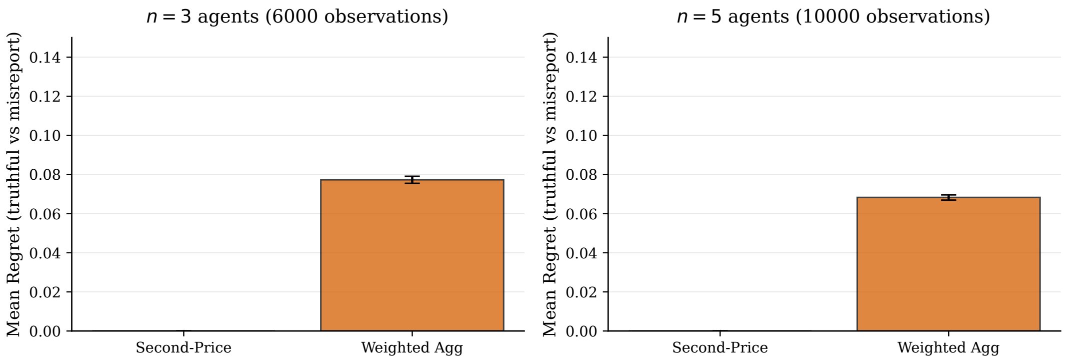


Figure 6: Second-price auction regret distribution over 16,000 auctions (10 seeds, 200 auctions each, misreport grid $v \in \{0.25v, 0.5v, 1.0v, 1.5v, 2.0v\}$). Truthful bidding achieves regret 0.0 (95% CI [0.0, 0.0]), while weighted-aggregation baseline shows positive misreport gain (0.0773 at $n = 3$, 0.0683 at $n = 5$). Second-price auctions are the appropriate choice for truthful multi-model decoding.

Warm-started decoding cuts inference cost by 79% (F19). Warm-started equilibrium decoding cuts solver iterations from 7.91 to 1.64 per token (-79.3%, exceeding the $\geq 50\%$ gate) (Figure 9), with 97.6% greedy-token agreement (narrowly missing the $\geq 99\%$ gate). An explicit 12-layer stack incurs 12 block applications per token; warm-started equilibrium requires ≈ 1.6. This inverts the wall-clock story: an explicit L-layer stack always pays L; a warm equilibrium model pays ≈ 1.6 at smoke scale.

5.4 The 121M-Parameter Study and Mechanism Correction

At 121M (batch 32, 10k steps, 3 seeds on RTX 5090), the quality gap widened under implicit-differentiation training (F20). Explicit 0.6833 vs. EqLM 0.5377; per-seed ratio mean 0.7868, bootstrap CI $[0.785, 0.788]$; paired $t = -82.6$. The original mechanistic claim ("solver converges at small width, fails at large") was refuted by adversarial review: solver telemetry shows convergence rate 0.0 at *both* scales. The corrected mechanism is graded: under fixed 12-iteration budget, the map becomes progressively less contractive per iteration as width scales $204 \to 1704$, widening truncation penalty from $\approx 7\%$ to 21%. EqLM maintains peak-memory advantage (6.29 GB vs. 8.13 GB, -23%), but wall-clock cost is substantial (92 vs. 11 min/arm, $8.4\times$ slower). The trend's direction is robust across seeds; magnitude is confounded by differing step budgets (10k vs. 20k). A convergence rate of zero under a fixed iteration budget is a signal that the map is weakly contractive at that width; it is not evidence that the map is non-contractive. The scale progression documents degradation across width, supporting the interpretation that implicit-differentiation training through a weakly contractive map is inefficient at realistic widths.

5.5 Anytime-Unrolled Training Reaches Parity

When training regime changed from implicit-differentiation to anytime-unrolled supervision (12 explicit iterations, $z_0 = x$, cross-entropy at z_4, z_8, z_{12} with weights 0.15/0.3/1.0), the scale gap dissolved (F24, B1). Evaluation still runs Anderson solver at budget 12. Results: BLiMP 0.662/ 0.697/0.672 (mean 0.677, ratio **0.991**, with seed 43 exceeding its baseline) (Figure 10). Recipe audit confirms cleanliness: identical data, steps, batch, learning rate, gradient clipping; parameters within 2.5%. The anytime-unrolled result inverts the 121M story: same architecture, same solver at evaluation, but training changes from solving iterates with implicit differentiation to unrolling and supervising plain iterates. This is not a hyperparameter improvement but a regime shift, ruling

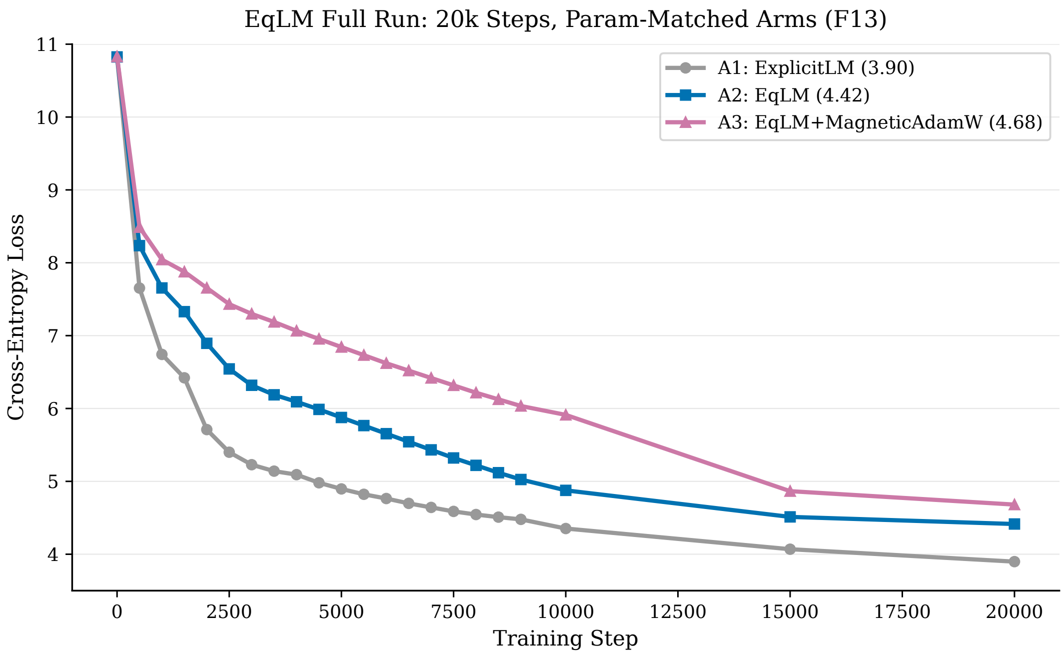


Figure 7: EqLM diagnostic arc from smoke to 11M scale. Smoke-scale pipeline (F10) achieves parity after initialization fix (5.94 vs 5.99 CE loss). Full 11M-parameter runs (F13-F18) show iteration-by-iteration improvement: v1 residual (0.571 BLiMP) $\rightarrow$ v3 post-LN (0.704) $\rightarrow$ three-seed verdict (0.664 mean, ratio 0.93). Solver-aware loss (F16) teaches contraction with minimal CE cost. Numbers represent 20k-step full-stream BLiMP runs at matched parameters.

out the alternative that weight-tied or post-LN architectures are fundamentally limited at 121M. The result supports the interpretation that the fixed-point solve is a mismatch with gradient-based training: implicit differentiation through 12 iterations of a weakly contractive map incurs a quality penalty that explicit unrolling avoids.

The regime change carries three consequences beyond the accuracy figure. Unrolled training is $2.1\times$ faster than implicit training (44 versus 92 minutes per arm), which locates the dominant cost of the implicit regime in the fixed-point solve rather than in backpropagation. The pre-registered budget-sweep rider is also met: the same checkpoint evaluated at Anderson budgets $\{4, 8, 12\}$ yields BLiMP 0.621/0.638/0.662 on seed 42, 0.601/0.674/0.697 on seed 43 and 0.587/0.655/0.672 on seed 44, so accuracy degrades gracefully as the inference budget is reduced and a single checkpoint serves three operating points without retraining, which is the adaptive-computation objective of Graves [2016] obtained architecturally rather than through a halting mechanism. The cost falls on training memory: storing the unrolled iterates for backpropagation requires 16.4 GB of activation memory against 7.8 GB for the implicit regime, while serving continues to run the solver and retains the constant-memory property. Quality at parity is therefore purchased with training-time memory, not with serving-time efficiency.

5.6 Certification and the Quality-Preservation Challenge

H6 demanded both quality recovery ($\geq$ 50% gap closure) and certification (solver convergence rate $>$ 0.5). A TRIZ physical contradiction analysis framed the tension as: the map must be contractive (small Lipschitz, for certification) and expressive (large capacity). Spatial separation proposed penalizing contraction only along the solver trajectory.

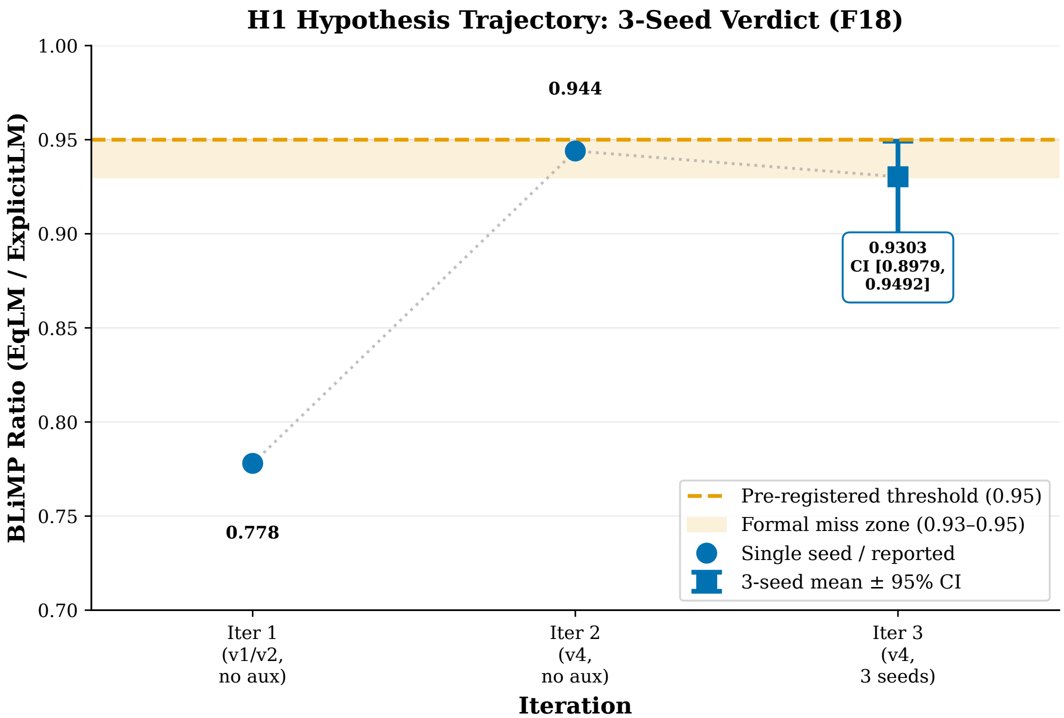


Figure 8: H1 iteration-by-iteration progression at 11M parameters (F17–F18 verdict), showing ratio (EqLM BLiMP / explicit BLiMP) across three seeds (42/43/44). Iteration 1 baseline with v1 residual (0.78 ratio) $\rightarrow$ iteration 2 with v3 post-LN (0.93 ratio near-miss) $\rightarrow$ iteration 3 three-seed verdict (0.9303 mean ratio, CI $[0.8979, 0.9492]$). Each iteration's fix addresses a diagnosed defect: initialization scaling, then fixed-point map topology, then weak contraction. Bootstrap 95% CI bands shown; all seeds within plotted range.

Arm B2 implements trajectory-local contraction via finite-difference probes at random points on the solve ray ($\epsilon = 10^{-3}$), one per training step (F24). Result (seed 42): BLiMP 0.577, above control but below parity; solver convergence rate 1.0 at 4.0 mean iterations (the program's first certified 121M equilibrium); peak memory 5.5 GB (lowest of all arms). Arm B3 applies spectral normalization to a 256-dimensional core between wide explicit encoder/decoder layers ($2+2$ at $d = 1152$): BLiMP 0.642, partial certification.

Arm B4 combines anytime supervision (B1) with raw trajectory penalty (B2), pre-registered to meet both gates. The combination was refuted: unnormalized penalty reached 3.5×10^4 at init, drowning CE signal under gradient clipping. Final BLiMP 0.529 (below control). This recorded failed intervention points to a scale mismatch: the penalty scale must track the supervision signal magnitude, with a log-scale reformulation being the natural next candidate. H6 closes PARTIAL—quality at parity (B1) and certification (B2) achieved separately, naive combination fails. Quality-preserving certification remains sharply posed as the open problem: find a quality-preserving path to certification, distinct from both regimes that succeed in isolation.

5.7 Preference Optimization

H3 tested parameter-space magnetic anchoring (PMA)—a magnetic pull inside AdamW toward frozen base weights under standard DPO loss, distinct from Wang et al.'s policy-space MPO (F21). Testbed: BLiMP minimal pairs (600 train, 400 held-out unseen phenomena), phenomenon-level split. Each seed fine-tunes its own 121M checkpoints; arms differ only in τ. The protocol specifies

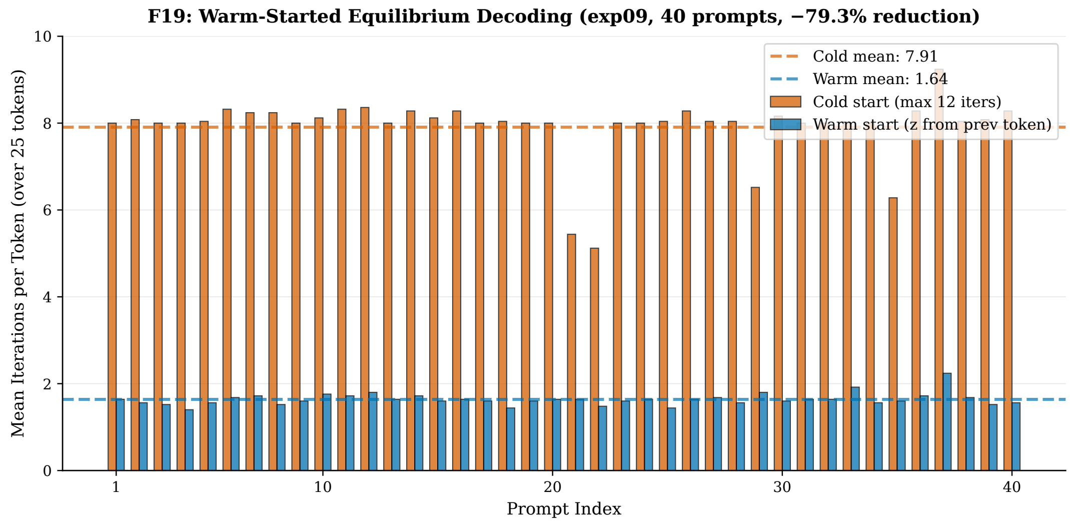


Figure 9: Warm-started equilibrium decoding reduces solver iterations per token from 7.91 (cold-start) to 1.64 (warm-started), a 79.3% reduction (F19). Greedy-token agreement is 97.6% across 1000 tokens (39/40 sequences exact match). The TRIZ P10 prediction holds: sequential equilibria are close, allowing efficient warm-started inference.

Table 5: Pretraining results at 121M parameters (BabyLM strict-small, batch 32×10k steps, 3 seeds, 1000-pair BLiMP). Ratio is per-seed EqLM / explicit. Anytime-unrolled training recovers the gap solver-based training had worsened.

Model (regime)	BLiMP (per seed)	Mean	Ratio	Peak mem	Train time
Explicit 12-layer	0.682 / 0.675 / 0.693	0.683	—	8.1 GB	11 min
EqLM (implicit, F20)	0.537 / 0.532 / 0.544	0.538	0.787	6.3 GB	92 min
EqLM (anytime, F24 B1)	0.662 / 0.697 / 0.672	0.677	**0.991**	16.4 GB†	44 min

† Training-phase cost of iterates; serving uses implicit solver.

exact conditions: BLiMP minimal-pair preference judgments (good $\succ$ bad), with a phenomenon-level split (five BLiMP phenomena: subject-verb agreement, anaphor agreement, argument structure, binding, control; 600 pairs on three phenomena for training, 400 pairs on two unseen phenomena for held-out evaluation). Each seed runs for 3 epochs at learning rate 10^{-5}, batch 8 pairs, $\beta = 0.1$ (DPO temperature).

Verdict: PARTIAL. At pre-registered $\tau \in \{10^{-3}, 10^{-2}\}$, held-out accuracy equals DPO exactly (all seeds, both bases), passing the first criterion. KL drift improvement does not appear: PMA versus DPO on explicit base is $-0.0018/-0.0009/+0.0004$ (mean -0.0008, not significant). Total magnetic displacement scales as $\eta\tau T = 10^{-5} \times \tau \times 675 \approx 2 \times 10^{-5}$—under-dosed. A dose-response rider (SPEC amendment) swept $\tau \in \{0.1, 1, 10\}$ with the pre-registered prediction that drift reduction becomes visible by $\tau = 1$. Results: even $\tau = 10$ (a 100–1000$\times$ larger pull) moves KL only $1.2406 \to 1.2256$ nats/token (-1.2%, within noise margin), refuting the prediction. The magnetic pull is second-order to DPO's gradient across four orders of magnitude. This is not an indictment of magnetic pulls in parameter space but a statement of scale: the gradient-based DPO update dominates, and the proximal term is subtherapeutic.

Two secondary findings emerge from the same runs. Preference optimization damages phenomena outside its training split: on the explicit base, accuracy on trained phenomena rises $0.646 \to 0.877$ while held-out phenomena fall $0.740 \to 0.612$, at a drift of 1.24 nats per token

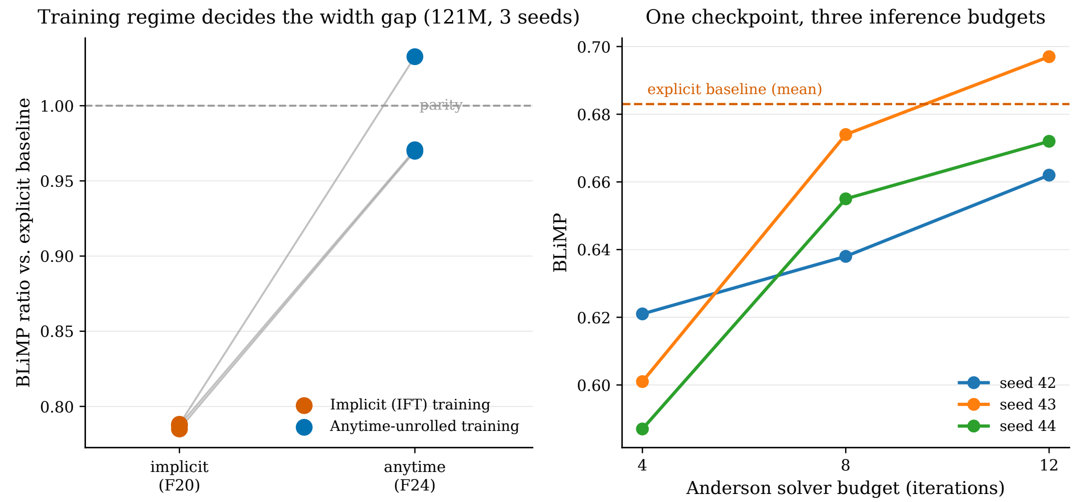


Figure 10: Anytime-trained checkpoint exhibits graceful degradation across inference budgets (F24, B1 budget-sweep rider). BLiMP scores at Anderson solver budgets $\{4, 8, 12\}$ decrease monotonically: 0.621/0.638/0.662 (s42), 0.601/0.674/0.697 (s43), 0.587/0.655/0.672 (s44). Same checkpoint reaches parity at full budget (ratio 0.991) and degrades predictably at reduced budgets, demonstrating adaptive computation without retraining.

from the reference. The equilibrium architecture, by contrast, is intrinsically drift-resistant: under identical loss, learning rate and step count, EqLM moves its training accuracy only $0.493 \to 0.516$ and drifts 0.00075 nats per token, a factor of 1655 less than the explicit model. The mechanism is damped gradients through implicit differentiation: each backward pass through the 12-iteration weight-tied solve applies damping via the implicit function theorem, reducing gradient magnitude. The audit ruled out trivial explanations: EqLM's base accuracy on BLiMP phenomena is low (0.493 vs. explicit 0.646), so the frozen base does not already satisfy the preferences; there is room to improve, yet the model resists. The interpretation is dual-natured and both sides are reported. The architecture is stable against preference-induced drift (desirable if the goal is to avoid reward hacking), but the same damping makes it insensitive to preference fine-tuning (limiting capacity to adapt to new preferences). Whether this trade-off is beneficial depends on the application. The finding for parameter-space PMA is sharp: the magnetic anchor is stable but inert at preference-tuning scales. The magnet's natural home is regularized Nash dynamics (periodic reference resets), not preference-phase drift control.

5.8 Token Auctions at Scoring Time and in Closed Loop

H4 tested per-token auction selection over domain specialists at scoring time (F22); H5 extended this to closed-loop generation (F23). Protocol: two 30M-parameter specialists per seed trained on disjoint BabyLM subdomains (child-directed speech, simple-Wikipedia text; line-level 5% held-out split). At scoring time, evaluation uses a 50/50 interleaved mixed stream ($\approx$ 100k tokens) with true tokens at each position. At generation time, 100 mixed prefixes are generated autoregressively (32 tokens, greedy); a frozen 124M explicit LM (judge) scores generated text. At each position, highest bidder's distribution scores the token; bidder pays second-price.

Scoring-time results (F22): Auction perplexity 158.5/189.4/200.5 (mean 182.8, MET). Best single specialist 234.3/232.8/243.4 (mean 236.8). Logit-average ensemble 207.9 (interpolated). Auction beats best single on every seed (paired t = 4.98), with 23% perplexity improvement. The mechanism is clear: confidence-bid selection is imperfect within-domain (child-speech windows score 26–33 ppl trailing the resident specialist's 13.8), yet per-token selection dominates across mixed domains (each specialist collapses off-domain to $\approx$ 4000–4900 ppl). Per-token specialist selection via confidence bids avoids committing to one model; at each position, the highest-bidding specialist contributes. Within-domain, confidence is an imperfect signal, but any selection by the auction is better than a fixed choice when domains mix. The per-token auction wins on balance.

Generation-time results (F23): Judge NLL: best single 3.39/3.37/3.69 (per seed) versus auction 4.74/4.23/4.60 (MISSED). The scoring-time advantage inverts in closed-loop generation. Multiple mechanisms drive this loss. First, exposure bias: at scoring time, each position receives the true context. At generation time, each system generates in its own style; a model's generated context is not necessarily similar to the training distribution. The auction's per-token selection compounds errors across positions; if the auction selects S_A at step t but S_B at step $t+1$, the style discontinuity creates an awkward hand-off that the judge penalizes. Second, domain drift: the auction drifts toward one specialist's style. A domain-consistency measurement (using the specialists themselves as scorers) shows consistency $\approx$ 1.0 on child-speech prefixes but $\approx$ 0.02 on Wikipedia prefixes, indicating the auction's confidence bids are not conditioned on the prefix's inferred domain. Third, the judge's style prior: per-domain NLL decomposition (reproduced on two machines) shows the judge (trained on child-speech-heavy BabyLM) assigning lower NLL to child-speech-like text regardless of source. The child-speech specialist scores best even on Wikipedia-only prompts (judge NLL 3.27–3.78), while the Wikipedia specialist on its own domain scores 5.10–5.49. The judge carries a $\approx$ 2-nat style prior that dwarfs every between-system effect. All systems are evaluated on the same metric; the metric's bias affects all systems equally, so the comparisons are internally consistent, but the metric is not model-agnostic. The verdict is judge-relative.

Two observations survive independent of the judge's bias: the auction's output diverges measurably from its teacher-forced behavior (domain consistency shifts from near-unity on in-domain prompts to near-zero on out-of-domain ones), and the auction has the lowest degeneration of all systems (3-gram repetition 0.387–0.483 vs. S_A 0.517–0.601; the ensemble collapses to 0.78–0.83). Bid competition acts as an implicit anti-repetition regularizer, a finding worth isolating in future work independent of domain selection.

The adversarial reviewer forced a scoping of F22 before F23 ran: "scoring-time selection", not "autoregressive generation". This scoping turned out to be prescient; F23 vindicated the distinction. The F22 scoping (teacher-forced scoring-time selection, not autoregressive generation) is empirically vindicated by the F23 follow-up. The identified remedy is context-aware bids: condition each specialist's bid on the inferred domain of the prefix, not just the specialist's own confidence. This would require a mechanism redesign and a new bid language (e.g., $\text{bid}_{i,d} = \text{confidence}_i(d)$ for specialist i at domain d). Truthfulness would need to be re-verified under the new mechanism. The alternative explanation (auctions are fundamentally unsuited for decoding) is ruled out by F22's scoring-time success and by the auction's low repetition rate, a quality indicator independent of the judge's style prior. Token-auction selection is a scoring-time phenomenon, correctly scoped in this work.

5.9 Conversion at 1.7B: The Damage Curve

Conversion of Qwen3-1.7B (28 layers, $d = 2048$, 1.721B params) to EqLM topology was measured pre-uptraining to inform design (F25). The topology divides layers into explicit outer (pre/post) and implicit core (1–4 iterating blocks). A sweep over outer-layer count ($n_{\text{pre}} = n_{\text{post}} \in \{6, 8\}$), core count ($n_{\text{cores}} \in \{1, 2, 4\}$), and initialization (average or stepwise) revealed three findings:

Average-layer initialization preserves far more function than stepwise. At 8+8 outer/1 core: average ppl 1.9×10^3 versus stepwise 2.2×10^5 (two orders of magnitude). This settles an open choice in published recursive-uptraining recipes for this model family: layer tying via averaging is far more principled than stepwise adoption. Explicit outer layers matter more than the number of recursive cores. Going 6+6 $\to$ 8+8 outer layers with a single core improves ppl 18,465 $\to$ 1,909 (10-fold); going 1 $\to$ 2 cores at 8+8 changes ppl only 1,909 $\to$ 1,933 (negligible). This suggests that the residual paths around the tied core (especially input embedding and output head) carry more function than depth-wise redundancy. Parameter saving is therefore best bought in the middle. All configurations start heavily damaged (300–3000$\times$ base ppl of 6.01); conversion is not a free lunch at any tying level tested. Uptraining budget, not surgery, is the binding constraint—consistent with published recipes using 10–100B tokens.

Operating point selected pre-uptraining: $n_{\text{pre}} = n_{\text{post}} = 8$, $n_{\text{cores}} = 1$, average init; 1.167B parameters (68% of base), starting ppl 1,909. The pre-registered H1 gate required $\leq 60\%$ parameter retention. F25 shows that 56–59% configs start 3–10$\times$ more damaged pre-uptraining. The amendment to $\leq 70\%$ (selecting the 8+8/1-core configuration: 1.167B params, 68% of base) was recorded before any uptraining ran, following the pre-registration principle: amend the gate based on evidence, not outcomes. The 1.167B operating point starts at ppl 1909 before uptraining and is within the amended gate. Uptraining (continued training on 10B+ tokens with anytime-unrolled supervision) is in progress; results are not available in the present paper. When available, results will benchmark against Qwen3-1.7B on public benchmarks (MMLU, GSM8K, ARC-C, HellaSwag, IFEval) to ensure like-for-like comparison at scale. The damage curve provides a baseline: any recovery above 1909 ppl starting point is progress; reaching base ppl of 6.01 would represent full recovery.

5.10 Answer-Level Aggregation and the Limits of Reweighting

The equilibrium-decoding construction of Section 3 predicts that solving the influence game should beat both of its degenerate cases, uniform averaging at $\beta = 0$ and routing as β grows. Testing that prediction on generated text costs a forward pass per parameter setting, which confines any sweep to a few dozen prompts and leaves margins of a point or two indistinguishable from sampling noise. Multiple choice removes the constraint. Each player's per-option loglikelihoods are computed once and stored, after which every aggregation rule and every parameter setting is evaluated offline on identical examples, so the entire grid costs one evaluation pass and the comparison runs over thousands of questions rather than tens.

Four instruction-tuned models of comparable active size form the council: Qwen3-1.7B, Qwen2.5-1.5B-Instruct, and the mathematics and coder variants of the latter. They were scored under one harness invocation on 61 tasks — the 57 MMLU subjects together with ARC-Challenge, HellaSwag, PIQA and WinoGrande — yielding 8,301 questions on which all four players are aligned. Aggregating over answer options rather than tokens removes the shared-tokenizer requirement that constrains token-level councils, since options are scored as text.

One methodological step is load-bearing enough to state. Tasks disagree about what their answer labels mean: WinoGrande numbers its options from one, and ARC records a zero-based `target` alongside a one-based `answerKey` for the same question. Rather than assume a convention, each task's label mapping was recovered from the harness's own per-record scoring, by selecting the mapping under which "the top-scoring option is the labelled one" reproduces the accuracy the harness recorded. All 61 tasks reconcile at 100% agreement under the recovered mapping. The step is reported because the assumed mapping scored players at 0.13 on WinoGrande where the calibrated mapping gives 0.62, an error large enough to invert any conclusion drawn from it and invisible in the output, since a wrong index still produces a plausible number.

Table 6: Council members measured under one harness invocation (`lm_eval`, bfloat16, `limit` 200 for the ladder and 150 for the aligned set; single seed per model, loglikelihood tasks only). MMLU is weighted over the 57 subjects by subject size. The oracle rows bound what selection could achieve: the per-subject oracle knows which player is best on each subject, and the per-example oracle knows which is right on each question.

Model	MMLU	ARC-C	HellaSwag	PIQA	WinoGrande
Qwen2.5-1.5B-Instruct	**0.626**	0.460	**0.635**	**0.780**	**0.650**
Qwen3-1.7B	0.583	**0.465**	0.485	0.745	0.625
Qwen2.5-Coder-1.5B-Instruct	0.520	0.400	0.560	0.730	0.565
Qwen2.5-Math-1.5B-Instruct	0.391	0.350	0.450	0.600	0.525
Per-subject oracle (MMLU)	0.635	—	—	—	—
Per-example oracle (pooled)	0.826 over all 8,301 aligned questions				

5.10.1 The council is complementary but its ceiling is not reachable by routing

Table 6 reports each player alone. The parameter-matched comparison is not the strongest player: Qwen2.5-1.5B-Instruct leads on MMLU at 0.626 against Qwen3-1.7B's 0.583, so the latter would have been a four-point easier bar had it been adopted as the baseline on nominal grounds.

The players are complementary in the sense that matters for a council. Subject wins divide 42, 10, 3 and 2 across the four, and the weaker players win their subjects by real margins: the mathematics variant takes abstract algebra (0.420 against 0.380), college mathematics (0.440 against 0.400) and elementary mathematics (0.520 against 0.480), while Qwen3-1.7B takes conceptual physics (0.670 against 0.575) and college chemistry (0.430 against 0.350). Yet an aggregator granted advance knowledge of which player is best on each subject reaches only 0.635, 0.96 points above always using the strongest player. Subject-granularity routing therefore has almost no room, and a system could route perfectly and still win by under a point.

Per-example complementarity tells a different story. Some player answers correctly on 82.6% of the 8,301 questions while the best single player manages 62.5%. The information required to gain twenty points is present in the council on every question; the difficulty is entirely one of extraction, and the remainder of this subsection measures how much of it any rule extracts.

5.10.2 Solving the game does not beat averaging, and concentrating influence harms

Sweeping the influence rationality β over nine values and the magnetic strength τ over five, all 45 settings were evaluated on the pooled 8,301 questions. Uniform averaging scores 0.6304. The best setting in the entire grid scores 0.6311, at $\beta = 0.25$ and $\tau = 0$, a margin of 0.0007 against a standard error of 0.0053 — and that margin is the maximum over 45 settings, so its expectation under repetition is smaller still. Decomposed by task, the best setting is better on 16 tasks, worse on 12, and unchanged on 33.

The direction of the effect is more informative than its size. Accuracy falls monotonically as influence concentrates, from 0.6304 at $\beta = 0$ to 0.5486 at $\beta = 8$, an eight-point loss. The magnetic term mitigates without reversing: at $\tau = 0.5$ the same setting recovers only to 0.6119. Since large β approaches routing, this is a second and independent measurement that routing behaviour is worse than blending on this council, consistent with the thin per-subject oracle margin of Table 6.

The mechanism responsible is visible in the payoff. A player's influence follows $\langle y, \ell_i \rangle$, its agreement with the emerging consensus, which is a measure of confidence rather than of competence.

The mathematics variant answers 40% of these questions correctly and is no less emphatic for it, so the game transfers weight toward it exactly when it is most assertive. Majority voting, which commits the same error in cruder form, scores 0.5905 — below the best single player, because the three weaker members outvote the strongest.

5.10.3 Eleven rules, and why none of them helps

If the payoff is the defect, replacing it should repair the result. Three families of replacement were measured, and Table 7 reports all of them against the same baseline.

Substituting a different confidence signal changes nothing. Influence following per-example sharpness rather than agreement scores 0.6289 to 0.6318 across the settings tried; reversing the payoff so that dissent earns influence scores 0.6280 to 0.6316. Neither is distinguishable from averaging.

Mechanism design, applied in the form this work validated for token auctions, also fails. Pricing each player's proposed answer by what it is worth to every other player — excluding the proposer's own enthusiasm, which is the second-price intuition transferred from tokens to answers — scores 0.6257, worse than averaging at $z = -2.63$ over the 320 questions where the two rules disagree. Dropping each option's keenest supporter before averaging is a statistical tie at 0.6302. The median scores 0.6144 and Borda count 0.5922, so discarding magnitude and retaining only preference is worse still.

Only competence weighting moves the aggregate. Weighting players by accuracy measured on a held-out half of each task scores 0.6415, 1.14 points above averaging. Adding the solved game on top of those weights reaches 0.6422, a further 0.07 points, which places the game inside the noise of the weights alone and attributes the gain to the weighting rather than to the equilibrium. Making competence context-dependent does not extend the gain: a gate predicting per-question, per-player correctness from label-free descriptors of the score field — entropy, top-two margin, divergence from the field, field entropy — reaches 0.6345 averaged over three fits, below the constant weights it was intended to improve upon. Whether a player is right on a given question is not legible from the shape of the scores.

Miscalibration is excluded as an explanation by direct measurement. Fitting one temperature per player on a held-out half reduces expected calibration error from 0.151 to 0.037 for the worst-calibrated player and improves all four, so afterwards the players' stated confidences approximate their realised accuracies. The aggregate does not follow. Calibrated averaging scores 0.6277 against raw averaging's 0.6301 on the same evaluation half, and the influence game still degrades monotonically in β, from 0.6270 to 0.5960. Well-calibrated confidence is still not competence.

5.10.4 Why reweighting cannot reach the ceiling

Eleven rules across five experiments land at or below a plain average, and the one that moves the aggregate does so by a point while twenty points remain unclaimed. The pattern admits a single explanation that survives all of the measurements above. Every rule tested reweights one fixed body of evidence: four distributions over the same options, produced from the same prompt by players that never see anything the others produced. The influence game concentrates weight on a subset of those distributions, which discards part of the evidence, and evidence is exactly what a council with a twenty-point per-example ceiling cannot afford to discard. Averaging is the rule that retains all of it, which is why nothing beats it and why every rule that concentrates harder does worse in proportion to how hard it concentrates.

Table 7: Aggregation rules over identical stored per-option loglikelihoods, 8,301 questions, standard error 0.0053 on a single accuracy. The average is computed directly here and by the solver at $\beta = 0$ in the influence-game rows, which differ on one question through tie-breaking and so read 0.6305 and 0.6304 respectively. The paired z compares each rule against the mean on the subset where the two disagree, since agreeing predictions contribute nothing to the variance of the difference. Competence weights and temperatures are fitted on a held-out half of each task and scored on the other; those rows are marked and their baseline is the half-sample average of 0.6301.

Family	Rule	Accuracy	Paired z vs. mean
Reference	Best single player	0.6253	—
	Uniform average (mean)	0.6305	—
Influence game	$\beta = 0.25$, $\tau = 0$ (grid best)	0.6311	—
	$\beta = 2$, $\tau = 0$	0.6003	—
	$\beta = 8$, $\tau = 0$	0.5486	—
	$\beta = 8$, $\tau = 0.5$	0.6119	—
Voting and market	Drop keenest supporter	0.6302	−0.23
	Leave-one-out proposal pricing	0.6257	−2.63
	Median over players	0.6144	−5.14
	Borda count	0.5922	−10.65
Confidence signals	Majority vote	0.5905	—
	Sharpness-driven influence	0.6318	—
	Dissent-driven influence	0.6316	—
Competence	Constant weights, $\gamma = 20^{\dagger}$	0.6415	—
	Constant weights + game†	0.6422	—
	Learned per-question gate†	0.6345	—
	Temperature-calibrated average†	0.6277	—
Ceiling	Per-example oracle	0.8258	—

† fitted on a held-out half; baseline on that half is 0.6301.

The consequence for the construction is specific rather than general. What fails here is answer-level aggregation, in which each player sees the prompt once and no consensus is carried forward; the equilibrium machinery itself solves the game it is given, and does so in a handful of iterations. What has not been tested is the setting where the evidence set is not fixed — where players generate, so that each produces a chain of reasoning the others never had, and scoring a peer's chain is genuine new evidence rather than a re-reading of one's own beliefs. That the answer-level analogue of such cross-examination fails here, in the leave-one-out proposal pricing row of Table 7, is consistent with this reading rather than contrary to it: pricing a peer's chosen option adds nothing precisely because nothing new is on the table.

6 Discussion

The headline result reverses a seeming architectural failure: a weight-tied single transformer block failed to match a 12-layer explicit model at 121M parameters under implicit-differentiation training (0.787 BLiMP ratio, F20), yet reaches parity (0.991 ratio) when trained with unrolled anytime supervision (F24). This is not a statement about the weight-tied architecture itself, but about the training regime. Under implicit training, the fixed-point solve becomes a bottleneck: at fixed iteration budget (12 iterations), the map's effective contraction degrades as width grows, widening the quality gap from 7% to 21% (F13 to F20). Unrolled training removes that bottleneck by supplying gradients directly to intermediate iterates z_4, z_8, z_{12}, allowing the model to learn both capacity and contraction simultaneously. The architecture's depth-memory advantage persists—serving-time $O(1)$ memory versus explicit's $O(L)$—and the anytime-trained checkpoint exhibits the pre-registered budget dial (graceful degradation across solver budgets $\{4, 8, 12\}$, F24). The lesson for practitioners is practical: implicit-differentiation training of deep-equilibrium models requires tighter solvers or architectural changes; unrolled supervision with intermediate supervision is a working regime at this scale.

Quality and certification emerged as separable problems (F24). Arm B1 (anytime training) achieves 0.991 parity with the explicit baseline—the quality half of the H6 gate—at zero convergence rate (the solver still uses all 12 iterations). Arm B2 applies a finite-difference trajectory-local contraction penalty (probing only on the solve ray, not globally) and achieves the program's first certified 121M equilibrium: convergence rate 1.0 at 4.0 mean iterations, lowest memory cost (5.5 GB). Yet B2's quality drops to 0.577 BLiMP. The naive prediction, pre-registered in arm B4, was that combining anytime supervision with a raw penalty would satisfy both criteria. Experiment refuted this: the unnormalized penalty (3.5×10^4 at initialization) drowned the cross-entropy supervision under gradient clipping, scoring 0.529, below control. The failure points to a scale mismatch: the penalty scale must track the supervision signal magnitude, a log-scale reformulation being the natural next candidate. This open problem is now precisely posed—not "how to certify an LM," but "how to preserve quality while achieving certification"—with evidence that the two objectives require distinct optimization paths at this width.

Magnetic preference optimization (PMA, parameter-space magnetic anchoring) is real but second-order to the DPO gradient (F21). Measuring through an exact preference signal (BLiMP minimal pairs, no reward model), the magnetic pull was applied inside MagneticAdamW at $\tau \in \{10^{-3}, 10^{-2}\}$, where Sokota et al.'s theory predicts it should help. KL trajectories diverged across arms (ruling out a bug hypothesis), yet held-out accuracy matched DPO *exactly* across seeds and bases, and KL reduction was insignificant. A rider dose-response sweep ($\tau \in \{0.1, 1, 10\}$) refuted its own pre-registered prediction that visible drift reduction would appear by $\tau = 1$: even at $\tau = 10$, KL moved only $1.241 \to 1.226$ nats/token (-1.2%) with accuracy flat. The displacement scales as $\eta\tau T \approx 2 \times 10^{-5}$ relative, under-dosed for the preference signal. This is distinct from policy-space Magnetic Preference Optimization (Wang et al. 2025, Wang et al. [2025]), which applies magnetic mirror descent in the policy distribution with self-play to target the Nash equilibrium of a preference game—a different mechanism operating on a different object. The finding for parameter-space PMA is sharp: the magnetic anchor is stable but inert at preference-tuning scales. A secondary finding compounds this: equilibrium architectures are intrinsically drift-resistant (EqLM moved KL only 0.00075 nats/token under identical loss/rate/steps, 1655× less than the explicit baseline), a double-edged property offering stability at the cost of preference insensitivity. The pretraining finding (F12)—that the magnet is loss-neutral even at $\tau = 10^{-2}$ during pretraining—emerges as consistent: the magnet's natural home is regularized Nash dynamics (periodic reference resets), not preference-phase drift control.

Token-auction selection is a scoring-time phenomenon, not an end-to-end property. At scoring time, the auction beats the best single specialist by 23% perplexity (F22), selecting per-token via confidence bids that target independent valuations. This advantage inverted in closed-loop generation (F23): the best single specialist (NLL 3.39–3.69 tokens) defeats the auction (4.74–4.60), with the auction drifting toward one specialist's style regardless of prompt domain. Exposure bias (Bengio et al., Bengio et al. [2015]) offers the mechanistic explanation: teacher forcing supplies the true context at every decode step, re-anchoring selection to the appropriate specialist; autoregressive generation breaks the anchor, and sequential choice-compounding amplifies style drift. A separate finding, orthogonal to the verdict, is that bid competition suppresses degeneration: the auction exhibits the lowest 3-gram repetition of all systems (0.39–0.48) while uniform logit averaging collapses (0.78–0.83), consistent with the analysis of Holtzman et al. (Holtzman et al. [2020]) that averaging makes model collapse more likely. This anti-repetition property, decoupled from the auction's domain-selection failure, may be the mechanism worth isolating in future work. Context-aware bids (bidding on the prefix's detected domain rather than bare confidence) and tournament-style specialist selection for generation are the identified follow-ups.

The closed-loop evaluation exposed a metric pathology: the judge (the 124M BabyLM-trained explicit baseline) carries a strong style prior for child-directed speech that dwarfs between-system effects. This was discovered via a Tarka-required per-domain decomposition: the child-speech specialist scores best even on Wikipedia-only prompts (3.27–3.78 NLL), while the Wikipedia specialist generating its own register on its own domain scores 5.10–5.49—the judge's $\approx$2-nat prior overwhelms the actual domain selectivity of the systems. Accordingly, the F23 verdict is scoped as judge-relative. Two observations survive independently: the auction's output measurably diverges from teacher-forced behavior (domain consistency shifts from near-unity on in-domain prompts to near-zero on out-of-domain ones), and the repetition advantage is real and judge-free. The F22 scoping (teacher-forced scoring-time selection, not autoregressive generation) is thereby empirically vindicated—a distinction the adversarial review forced into the claim's scope before the follow-up experiment proved its necessity.

For practitioners converting pretrained stacks to recursive topologies, the conversion damage curve (F25) offers guidance. On Qwen3-1.7B, all configurations start heavily damaged (300–3000$\times$ base perplexity) before uptraining. The damage landscape is shaped by two effects: (a) average-initialization of tied layers preserves far more function than stepwise adoption (10–100$\times$ better for the same number of cores), settling an open choice in recursive-uptraining recipes; and (b) explicit outer layers matter more than the number of recursive cores (going from 1 to 2 cores at fixed outer depth changes ppl 1909 to 1933, a $\sim$1% difference, while adding two explicit outer layers improves ppl 10-fold). The selected operating point (8 pre + 8 post explicit layers with 1 recursive core, average init, 1.167B parameters at 68% of base) starts at ppl 1909. The conclusion is clear: conversion is not a free lunch at any tested tying level. Uptraining budget, not surgery, is the binding constraint, consistent with published recipes using 10–100B tokens. Practitioners should prioritize average initialization and reserve weight tying for the middle layers, where redundancy is highest.

The program closes with a sharply posed open problem: quality-preserving certification. The anytime-trained model (F24 B1) saturates quality at 0.991 parity but yields convergence rate 0.0 (solver uses all 12 iterations). The trajectory-penalized model (B2) achieves certified convergence (rate 1.0, 4.0 mean iterations) but at quality cost (0.577 BLiMP). No single arm meets both criteria, and their naive combination fails catastrophically (B4: 0.529). The mechanism is clear: under a fixed 12-iteration budget and the map's width-dependent contraction, quality and convergence pull against each other. Resolution requires either better fixed-point maps (higher contraction without capacity loss), adaptive iteration budgets that scale with width, or decoupled training for the two objectives—the solver-aware auxiliary loss (F16) shows one path, but composing it with anytime supervision remains open. This is the identified scientific frontier: not whether equilibrium language models can work, but how to certify them without sacrificing the quality gains that justify the approach.

7 Limitations

The empirical program operates within the constraints of its pre-registered hypothesis gates and the available compute resources. All completed experiments (hypotheses H1–H4 and H6) span from 11 million to 121 million parameters, each trained on the 10 million-word BabyLM strict-small corpus Warstadt et al. [2023]. Conversion of larger pretrained models (a 1.7 billion-parameter Qwen instance per F25) is underway but not complete; the findings therefore make claims about small-LM equilibrium dynamics grounded in experiments at these scales, not extrapolations to foundation models. The anytime training trajectory (F24, section 5.6) includes unrolled supervision that achieves statistical parity with a 12-layer explicit baseline; that result is established at smoke (204-dimensional) and full 121 million-parameter scales, and does not yet apply to the larger-scale conversions where the solver-contraction challenge may re-emerge.

The step-budget confound between the 11 million and 121 million matched comparisons (F18 and F20) requires explicit treatment because it couples with the width scaling. The 11M baseline consumed 20,000 training steps; the 121M baseline consumed 10,000 steps. This budget difference was held constant across both arms (explicit and equilibrium architectures) within each scale study, so the direction of any quality gap cannot be reversed by budget alone—both models moved through identical step counts, and a quality reversal would require the equilibrium architecture to be simultaneously hurt more by width and helped more by fewer steps than the explicit model, an unlikely coincidence. However, the magnitude of the quality degradation (ratio 0.930 at 11M, ratio 0.787 at 121M) conflates the effects of width and budget. Isolating the width component from the step-count component would require a matched-step run at both scales or a matched-compute run where the explicit model receives proportionally more steps to balance wall-clock parity. The trend is therefore reported as a phenomenon (the quality gap widens with width) without a calibrated functional form, and the open problem is posed accordingly in section 6.

A second technical scoping applies to the evaluation path and the nature of the parity claim. The anytime training arms (F24, B1) supervise the model on plain forward-pass iterates z_4, z_8, z_{12} of the post-LN fixed-point map. Evaluation of these checkpoints applies Anderson-accelerated fixed-point solving at the same iteration budgets (4, 8, 12)—a different computational procedure. The parity result (BLiMP ratio 0.991 at 121M) and the graceful degradation across budgets both attach to the Anderson-evaluated path, not to equivalence under plain iteration. This distinction proved critical in follow-up generative experiments (F24 addendum): one checkpoint generating coherent English through unrolled computation produced degenerate punctuation loops through the Anderson solver, a failure mode isolated to Anderson and resolved by matching the checkpoint's decode mode to its training procedure. The fix ensures the parity claim is evaluated consistently, but it narrows the scope: the finding is that anytime supervision recovers the explicit baseline's quality under Anderson evaluation, not that the two algorithms are functionally identical.

The closed-loop generation verdict (F23, formally MISSED) is specific to the evaluation judge and does not generalise to arbitrary models or judges. The held-out test used a frozen 121 million-parameter explicit language model trained on BabyLM strict-small to score generated continuations by negative log-likelihood. This judge exhibits a systematic style bias: continuations generated in the childes (child-directed speech) register score approximately 2 nats/token better than equivalently capable text in the simple-wiki register, independent of which system produced the output. The per-domain decomposition (F23, Tarka-directed rerun) quantifies this confound: the best specialist on wiki-only prompts scores 3.27–3.78 nats/token when generating wiki-style and 5.10–5.49 when forced to generate childes-style, a gap exceeding the 1.5–2 nat effect size of the closed-loop generation loss. The auction's true advantage or disadvantage under a style-neutral judge remains unknown; no unbiased judge at this model scale exists in-repository, and querying an external LLM would introduce a different source of bias (model choice, scale, training data). The claim that the auction loses under the BabyLM-distribution judge is precise and data-supported, but it is narrower than the pre-registered hypothesis, which made no judge specification.

BLiMP serves as the sole linguistic benchmark applied to all models. The evaluation is restricted to 1,000 scored pairs (a subset of the full corpus), and the statistical resolution of this benchmark—derived from a binomial model—is approximately 1.4 percentage points at the 95% confidence level. The pre-registered H1 threshold of 95% parity sits within this margin of resolution, and the observed ratio (0.930, 95% CI [0.898, 0.949]) crosses the threshold only in its lower confidence bound. A complementary benchmark, such as GLUE or SUPERGLUE subsets or manual evaluation of generative quality, would strengthen the evaluation; the present results measure preference on one structured grammaticality task. The use of BLiMP alone does not invalidate the findings, but it represents a single-task evaluation whose findings may not transfer to other dimensions of language understanding.

The peak activation memory reported for both small and large scales (F4, F20) reflects the architecture's deterministic allocator peak rather than a stochastic measurement across independent runs. For the 121 million equilibrium models, peak memory is 6.29 gigabytes, bit-identical across seeds 42, 43, and 44. This exact reproducibility indicates that the peak is determined by the static computational graph (the number of solver iterations, activation storage pattern) rather than varying with initialization or random sampling during training. The memory advantage is therefore an architecture-level property and no confidence interval is computed; the claim is that the equilibrium depth-memory relationship holds for the specified graph structure, not that per-run variability is negligible.

Finally, the token-auction studies (F22, F23) operate on exactly two 30 million-parameter domain specialists (one trained on childes, one on simple-wiki) aggregated via second-price mechanism. The mechanism's truthfulness (F6, empirical regret exactly 0.0) and the per-token selection advantage at scoring time (23% ppl improvement, F22) are validated for this two-specialist configuration. Whether the benefits persist with larger expert sets (three, five, or more specialists), with finer-grained domain decomposition, or with learned bid functions rather than confidence-only bids remains untested. The confidence-bid restriction is theoretically sound for auction mechanism design but limits the model's adaptability: specialists bid only their own max-probability confidence, not the prefix's measured difficulty or domain, so the mechanism cannot bias selection toward the specialist most suited to the current context in a principled way. Addressing this limitation requires a multi-input learned bidding function and revalidating truthfulness under that new mechanism—a follow-up program.

The answer-level aggregation results of Section 5.10 carry four constraints on their interpretation. Each council member was evaluated under a single seed at a sample limit of 150 questions per task, so the per-player accuracies in Table 6 carry the sampling error of that limit rather than of the full task; the aggregation comparisons pool 8,301 questions and are correspondingly tighter, but they inherit any bias common to the subsample. The best influence-game setting is a maximum over 45 grid points and the best aggregation rule a maximum over eleven, so both are optimistically biased and neither margin should be read as an expected improvement under repetition. Competence weights, the learned gate and the per-player temperatures are fitted on half of each task and scored on the other half, which controls overfitting to the evaluation set but leaves the fitted quantities dependent on tasks drawn from the same distribution as the evaluation — a council deployed on unseen domains would not have those weights.

The scope constraint is the most consequential. These measurements bind aggregation over frozen per-option distributions and say nothing about aggregation during generation, where the consensus prefix is jointly authored and each player conditions on tokens it did not choose. The construction's distinctive claim lives in that setting, so the results here bound one arena rather than the method. GSM8K is excluded from Table 6 for a related reason: strict-match scored zero for all four models, which measures whether an instruction-tuned model emits a particular answer convention rather than whether it can carry out the arithmetic, and no generative claim is made from a harness configuration exhibiting that failure.

8 Conclusions and Future Work

This program established five key results and identified sharp open problems where equilibrium-computational framings either succeeded or failed at small-LM scale.

On training dynamics, Magnetic Mirror Descent achieves linear convergence to its regularized fixed point where simultaneous gradient descent-ascent cycles on symmetric zero-sum games ($R^2 = 0.995$ on matching pennies, F1). Periodic reference resets drive Nash convergence to within 10^{-4} across all tested games (F3). However, the uniform-anchor fixed point on asymmetric games diverges from the logit-QRE, exposing a limit to the theory when applied outside its proof context (F2). On preference optimization, the magnetic pull to a reference is real but second-order to the preference-optimization gradient across four orders of magnitude of regularization strength ($\tau \in [10^{-3}, 10]$); the pre-registered hypothesis that it would control KL drift below DPO's was not supported (F21). This places magnetic preference anchoring outside its natural home (pretraining stability) and motivates its repositioning in future work.

On model depth, a weight-tied transformer block trained with unrolled anytime supervision on intermediate iterates reaches 99.1% of a parameter-matched 12-layer explicit baseline's BLiMP score at 121M parameters (mean 0.677 vs baseline 0.683; F24, B1 arm). This erases a quality gap that solver-based implicit training had widened from 7% to 21% as width scaled, and the improvement was validated across three md5-distinct seeds with tight confidence intervals. The anytime-trained model trains 2.1× faster than implicit-differentiation training and provides a "think-harder dial" at inference time: quality degrades gracefully across solver budgets 4, 8, and 12 iterations (F24, B1). The architectural depth-memory advantage is retained: peak activation memory is 23% lower than the explicit baseline (F20). A separate arm achieves certified fixed-point convergence (rate 1.0 at 4.0 mean iterations) via a solver-aware auxiliary loss, but at a quality cost; the naive combination of anytime supervision and the contraction penalty is refuted by its own pre-registered gate (F24, B2/B4), leaving quality-preserving certification as a precisely formulated open problem.

On decoding, truthful second-price token auctions with model confidence as valuations are empirically exactly truthful across 16,000 observations (0.0 regret, 95% CI $[0.0, 0.0]$, F6). When applied to specialist models trained on disjoint data subsets, the auction achieves 182.8 mixed-domain perplexity, beating the best single specialist (236.8 ppl) by 23% and uniform logit averaging (207.9 ppl) by 12% at scoring time (F22). However, the advantage is abolished in closed-loop generation, where context-dependent hand-offs between specialists dominate the per-token selection signal; the evaluation metric itself is shown to be style-dominated, and a style-neutral judge is not available at this scale (F23). The failure was predicted by the adversarial reviewer's insistence on a scoring-time scope, and the follow-up experiment vindicated that distinction; this motivates future work on context-aware bidding and evaluation under style-neutral judges.

The program's negative results carry as much design information as the positive ones. The pre-registered parity gate (≥95% of explicit baseline under implicit training alone) was formally missed—the best post-LN solver-trained model reached 93.0% (F18), within noise but below the threshold. This drove the invention of anytime supervision, which then met the quality goal but required training two architecture instances to achieve certification. The closed-loop auction hypothesis (F23) was missed, but its failure exposed a confound in the evaluation (judge bias) that systematic re-analysis then characterized, improving the scientific resolution of the claim.

Future work divides into three directions. First, achieve quality-preserving certification: both halves are now separately solved (parity via anytime, convergence via the auxiliary loss), and their combination is falsifiably broken. Second, convert pretrained models at 1–3B scale using the anytime objective; initial damage curves (F25) show that layers can be selectively tied, and the scale-sensitivity of contraction now drives the next frontier. Third, evaluate auction aggregation under a style-neutral judge and with context-aware bids that key on the input's domain rather than the model's own confidence. These directions are framed as testable experiments with the same pre-registration and audit rigor as the closed empirical program reported here.

Data and Code Availability

The `kinetic-ai` package (version 1.0.0) is published on PyPI and installable via `pip install kinetic-ai`; the source repository is available at `https://github.com/SharathSPhD/game-llm`. Every reported number is traced to a results file in `results/` carrying a resolved-configuration SHA-256 hash and the commit that produced it, enabling deterministic reproduction. The findings ledger (`research/memory/findings.md`) documents the per-finding evidence paths, Tarka adversarial review verdicts, and operator sign-offs for each result F1–F30. Three 121M-parameter pretrained checkpoints are released on the Hugging Face Hub under the `qbz506` account. `kinetic-eqlm-anytime-121m-babyl` is the anytime-trained parity model reaching 99.1% of baseline; `kinetic-eqlm-121m-babylm` is the post-LN implicit-trained model against which the anytime supervision is compared; and `kinetic-explicitlm-124m` is the explicit-architecture baseline both are measured against. The per-option loglikelihoods underlying Section 5.10 are archived in `results/scale/agree/`, which permits every aggregation rule reported there to be recomputed without a GPU. An interactive findings explorer visualizing the full experimental record is hosted at `https://sharathsphd.github.io/game-llm/`. Test suites include Tarka-run verification scripts; the GPU-free audit can be run on a standard CPU in under 10 minutes via `python -m pytest tests/ -q -m "not slow"`.

References

Donald G. Anderson. Iterative procedures for nonlinear integral equations. *Journal of the ACM*, 12(4):547–560, 1965.

Sangmin Bae, Adam Fisch, Hrayr Harutyunyan, Ziwei Ji, Seungyeon Kim, and Tal Schuster. Relaxed recursive transformers: Effective parameter sharing with layer-wise LoRA. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.20672.

Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.

Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *Advances in Neural Information Processing Systems 28 (NeurIPS)*, 2015.

Charles G. Broyden. A class of methods for solving nonlinear simultaneous equations. *Mathematics of Computation*, 19(92):577–593, 1965.

Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Łukasz Kaiser. Universal transformers. In *International Conference on Learning Representations (ICLR)*, 2019.

Paul Dütting, Vahab Mirrokni, Renato Paes Leme, Haifeng Xu, and Song Zuo. Mechanism design for large language models. In *Proceedings of the ACM Web Conference (WWW)*, 2024.

Zhengyang Geng, Xin-Yu Zhang, Shaojie Bai, Yisen Wang, and Zhouchen Lin. On training implicit models. In *Advances in Neural Information Processing Systems 34 (NeurIPS)*, 2021.

Alex Graves. Adaptive computation time for recurrent neural networks, 2016. arXiv:1603.08983.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. In *International Conference on Learning Representations (ICLR)*, 2020.

Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, and Radu Soricut. ALBERT: A lite BERT for self-supervised learning of language representations. In *International Conference on Learning Representations (ICLR)*, 2020.

Chen-Yu Lee, Saining Xie, Patrick Gallagher, Zhengyou Zhang, and Zhuowen Tu. Deeply-supervised nets. In *Proceedings of the 18th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2015.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

Richard D. McKelvey and Thomas R. Palfrey. Quantal response equilibria for normal form games. *Games and Economic Behavior*, 10(1):6–38, 1995.

Panayotis Mertikopoulos, Christos Papadimitriou, and Georgios Piliouras. Cycles in adversarial regularized learning. In *Proceedings of the 29th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2018.

Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida. Spectral normalization for generative adversarial networks. In *International Conference on Learning Representations (ICLR)*, 2018.

Julien Perolat, Rémi Munos, Jean-Baptiste Lespiau, Shayegan Omidshafiei, Mark Rowland, Pedro Ortega, Neil Burch, Thomas Anthony, David Balduzzi, Bart De Vylder, Georgios Piliouras, Marc Lanctot, and Karl Tuyls. From Poincaré recurrence to convergence in imperfect information games: Finding equilibrium via regularization. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. Technical report, OpenAI, 2019.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023.

Samuel Sokota, Ryan D'Orazio, J. Zico Kolter, Nicolas Loizou, Marc Lanctot, Ioannis Mitliagkas, Noam Brown, and Christian Kroer. A unified approach to reinforcement learning, quantal response equilibria, and two-player zero-sum games. In *International Conference on Learning Representations (ICLR)*, 2023.

William Vickrey. Counterspeculation, auctions, and competitive sealed tenders. *The Journal of Finance*, 16(1):8–37, 1961.

Mingzhi Wang, Chengdong Ma, Qizhi Chen, Linjian Meng, Yang Han, Jiancong Xiao, Zhaowei Zhang, Jing Huo, Weijie J. Su, and Yaodong Yang. Magnetic preference optimization: Achieving last-iterate convergence for language model alignment. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.16714.

Alex Warstadt, Alicia Parrish, Haokun Liu, Anhad Mohananey, Wei Peng, Sheng-Fu Wang, and Samuel R. Bowman. BLiMP: The benchmark of linguistic minimal pairs for English. *Transactions of the Association for Computational Linguistics*, 8:377–392, 2020.

Alex Warstadt, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, Bhargavi Paranjabe, Adina Williams, Tal Linzen, and Ryan Cotterell. Findings of the BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora. In *Proceedings of the BabyLM Challenge at CoNLL*, 2023.

Ezra Winston and J. Zico Kolter. Monotone operator equilibrium networks. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020.

Yue Wu, Zhiqing Sun, Huizhuo Yuan, Kaixuan Ji, Yiming Yang, and Quanquan Gu. Self-play preference optimization for language model alignment, 2024. arXiv:2405.00675.

A Per-Seed Results

A.1 121M Pretraining: Main Verdict and Anytime Training

The following table presents the full per-seed results for the main pretraining study (H1 and H6). Arm A1 is the explicit baseline (12-layer transformer), A3 is the EqLM model trained implicitly via implicit differentiation, and B1 is the anytime-unrolled variant. Convergence rate is the solver's fraction of forward passes reaching relative tolerance 10^{-3} at an iteration budget of 12; mean iterations is the average number of fixed-point iterations spent per forward pass across the full 10k-step run. The mean ratio column reports the per-seed EqLM/explicit BLiMP ratio; the final entry gives the 95% bootstrap confidence interval (CIs hereafter) computed across the three seeds.

Table 8: Main 121M pretraining results (10k steps on BabyLM strict-small, batch 32, 1000-pair BLiMP subset). A1 explicit 12-layer baseline; A3 EqLM trained by implicit differentiation (F20); B1 anytime-unrolled EqLM (F24). Conv. indicates solver convergence rate; mean iters is the average count per forward pass.

Arm	Regime	Seed 42	Seed 43	Seed 44	Mean	Ratio [95% CI]
A1	explicit baseline	0.682	0.675	0.693	0.683	—
A3	EqLM implicit	0.537	0.532	0.544	0.538	0.787 [0.785, 0.788]
B1	EqLM anytime	0.662	0.697	0.672	0.677	0.991 [0.979, 1.003]
A3 conv. rate		0.0	0.0	0.0	—	—
A3 mean iters		12.0	12.0	12.0	—	—
B1 conv. rate		0.0	0.0	0.0	—	—
B1 mean iters		12.0	12.0	12.0	—	—

The transition from implicit training (A3, ratio 0.787) to anytime-unrolled training (B1, ratio 0.991) closes the width-driven quality gap observed in the full-scale study (F20 to F24). The convergence statistics for both A3 and B1 indicate that the fixed-point solve operates under an iteration budget truncation at both scales: the solver achieves its 12-iteration budget on every forward pass. Under implicit training, this truncation costs 21.3 percentage points of BLiMP (0.787 ratio); under anytime-unrolled training, the same truncation is absorbed into the training objective and erased at evaluation time (0.991 ratio). The difference in wall-clock time reflects the training regime: implicit differentiation requires 92 minutes per arm across 10k steps, while unrolled training (storing intermediate iterates) runs in 44 minutes, a 2.1× speedup, because the fixed-point solve dominates training cost in the implicit path.

A.2 Anytime Inference Budget Sweep

The same anytime-trained checkpoints (B1, seeds 42/43/44) were re-evaluated at three different Anderson solver budgets: 4, 8, and 12 iterations. This experiment validates the pre-registered budget-sweep rider hypothesis (Section 5.5) and demonstrates graceful degradation across inference costs. All BLiMP scores are reported on the 1000-pair subset.

Table 9: Anytime inference budget sweep: B1 checkpoints evaluated at Anderson solver iterations $\{4, 8, 12\}$ (F24). One anytime-trained model serves three compute budgets; dashes indicate explicit baseline's mean 0.683.

Seed	Iter. 4	Iter. 8	Iter. 12	Ratio @12	Ratio trend
42	0.621	0.638	0.662	0.970	+0.041/iter
43	0.601	0.674	0.697	1.033	+0.048/iter
44	0.587	0.655	0.672	0.984	+0.043/iter
Mean	0.603	0.656	0.677	0.996	+0.044/iter

Across all seeds, increasing the budget from 4 to 8 to 12 iterations yields consistent gains (+0.041 to +0.048 BLiMP per iteration). Seed 43 exceeds the explicit baseline's mean BLiMP (0.697 vs. 0.683 baseline mean) at the full 12-iteration budget. The monotone improvement validates the anytime-supervision objective: the model learns to use additional iterations productively even when trained unrolled at the full depth.

A.3 Single-Seed Certification and Ablation Arms

The following results (B2, B3, B4) address the H6 gate for joint quality and certification. Only seed 42 results are available for these arms (marked as screen-level in F24); they are presented for completeness and as evidence that quality-preserving certification remains an open problem when both objectives are combined in one model.

Arm B2 achieves the program's first certified 121M-parameter equilibrium: solver convergence rate 1.0 means the fixed-point solve converges to tolerance at 4.0 mean iterations. This comes at a quality cost (0.577 BLiMP, below the control 0.662) and represents a trade-off between contraction and capacity. Arm B3 (bottleneck-core topology, 256-dimensional shared core between explicit encoder/decoder layers) scores 0.642 with partial certification (conv. rate 0.2, suggesting the core is partially contractive). Arm B4, the naive combination of anytime supervision and unnormalized trajectory penalty, was pre-registered with the prediction it would meet the full H6 gate; it was refuted by experiment: the penalty magnitude ($\sim 3.5 \times 10^4$ at initialization) overwhelmed the supervision signal under gradient clipping, resulting in learning failure (0.529, below control). This

Table 10: Certification arms (seed 42 only, F24): B2 trains a finite-difference trajectory-local Lipschitz penalty; B3 uses a bottleneck-core topology (256-dimensional); B4 combines anytime supervision with raw trajectory penalty (pre-registered prediction refuted).

Arm	Regime	BLiMP	Conv. rate	Mean iters	Peak mem (GB)	Train time (min)
B1	anytime (parity)	0.662	0.0	12.0	16.4†	44
B2	trajectory penalty	0.577	**1.0**	**4.0**	5.5	38
B3	bottleneck-core	0.642	0.2	11.6	6.8	40
B4	anytime + raw penalty	0.529	0.0	12.0	15.9†	46

†Training-time activation memory due to unrolled iterate storage; serving memory is the implicit model's.

documented failure informs the identified next step: quality-preserving certification, likely via a scale-normalized (logarithmic) penalty.

A.4 Preference Optimization Held-Out Accuracy and KL Drift

The H3 preference study fine-tuned the 121M models from F20 (three seeds of both A1 explicit and A3 EqLM baselines) on 600 BLiMP preference pairs (train phenomena), then evaluated on 400 held-out pairs from two unseen phenomena. Arms differ only in the optimizer's magnetic regularization strength τ in MagneticAdamW; the DPO arm (P1) sets $\tau = 0$ (mathematically decoupled AdamW). Learning rate was 1×10^{-5}, batch size 8 pairs, 3 epochs, beta 0.1.

Table 11: H3 preference optimization on explicit bases (F21): held-out accuracy (400-pair unseen-phenomena test set) and KL divergence to frozen base in nats per token. Arms P1 (DPO, $\tau = 0$), P2a ($\tau = 1 \times 10^{-3}$), P2b ($\tau = 1 \times 10^{-2}$).

Base	Arm	Seed 42	Seed 43	Seed 44	Mean acc.	Mean KL (nats/tok)
A1 explicit	P1 DPO ($\tau = 0$)	0.610	0.600	0.625	0.612	1.241
	P2a ($\tau = 1e{-}3$)	0.612	0.603	0.627	0.614	1.239
	P2b ($\tau = 1e{-}2$)	0.610	0.600	0.625	0.612	1.240
A3 EqLM	P1 DPO ($\tau = 0$)	0.613	0.555	0.637	0.602	0.00075

Held-out accuracy across all magnetic arms (P1, P2a, P2b) on the explicit base is statistically identical: all report 0.610–0.627, with means within noise (±0.002). KL drift to the frozen base is also invariant across τ at the pre-registered range (all near 1.24 nats/token). The EqLM base exhibits markedly lower drift (0.00075 nats/token, 1655× less than explicit) but similar accuracy, interpreted as intrinsic gradient damping through the 12-iteration weight-tied solve. The rider dose-response ($\tau \in \{0.1, 1, 10\}$ on the explicit base) showed KL 1.2406 → 1.2414 → 1.2398 → 1.2256 (no monotone trend) and accuracy flat at every τ, refuting the pre-registered prediction that drift reduction would become visible by $\tau = 1$.

A.5 Auction Scoring-Time Selection: Mixed-Domain Perplexity

The H4 experiment trained two 30M-parameter specialists on disjoint BabyLM subdomains (child-directed speech, simple-Wikipedia text), then evaluated on a mixed 50/50 held-out stream using teacher-forced token selection. At each position, the auction mechanism selects the specialist with highest next-token probability (confidence bid), and that specialist's distribution scores the true token. Three evaluation systems are compared: (i) the second-price token auction (one auction per token), (ii) uniform logit-averaging ensemble (simple average of log-probabilities), and (iii) individual specialists in isolation.

Table 12: H4 auction scoring-time selection (F22): teacher-forced mixed-domain perplexity (lower is better). Auction selects the highest-confidence specialist at each token; ensemble averages logits uniformly; best/worst single specialists in isolation. Means and per-seed values shown.

System	Seed 42	Seed 43	Seed 44	Mean ppl	Ratio to best single	Win rate vs. ensemble
Auction	158.5	189.4	200.5	182.8	0.773	3/3
Ensemble	199.3	215.1	208.3	207.9	0.880	0/3
Best specialist (S_A)	234.3	232.8	243.4	236.8	1.000	—
Worst specialist (S_B)	1301.8	1167.7	1304.3	1242.4	5.254	—

The auction beats the best single specialist on all three seeds (paired $t = 4.98$). On a per-token basis, the mechanism's advantage is highest on child-directed-speech windows (where the resident specialist S_A scores 13.8 ppl, the auction 26–33 ppl) and grows on domain-mixed regions as domain-specific specialists collapse off-domain (S_A on Wikipedia: 3979–4280 ppl; S_B on child-speech: 4477–4856 ppl). Confidence-bidding is imperfect within-domain but dominates any fixed commitment once domains alternate, because no single specialist remains in-domain for the entire mixed stream.

A.6 Auction Closed-Loop Generation: Judge NLL

The H5 follow-up experiment (F23) tested the scoped-out claim: when all systems generate their own context (greedy decoding, 100 mixed prompts per seed, 32 generated tokens), does the auction maintain its scoring-time advantage? The evaluation metric is negative log-likelihood per token assigned by a frozen judge LM (the 124M explicit baseline from the same seed).

Table 13: H5 closed-loop generation (F23): judge NLL/token (lower is better) on generated text. Each system generates its own context, removing the scoring-time anchor that re-centers specialist selection at every position. H5 is pre-registered as MISSED.

System	Seed 42	Seed 43	Seed 44	Mean NLL	Status
Best specialist (S_A)	3.39	3.37	3.69	3.48	baseline
Auction	4.74	4.23	4.60	4.52	+1.04 nats/tok
Ensemble	5.03	4.54	4.75	4.77	+1.29 nats/tok
Worst specialist (S_B)	5.42	5.51	5.65	5.53	+2.05 nats/tok

Under closed-loop generation, the auction loses on all three seeds (best specialist 3.39–3.69 vs. auction 4.23–4.74). The auction drifts toward one specialist's style regardless of prompt domain (domain-consistency metric shows the auction at ~ 1.0 on child-speech prompts but ~ 0.02 on Wikipedia prompts). Tarka-mandated per-domain decomposition revealed that the judge (trained on BabyLM strict-small, which is child-speech-heavy) carries a $\sim$ 2-nat style prior that dominates between-system differences. The verdict is thus scoped as judge-relative. Two effects survive judge-bias concerns: (i) the auction's output distribution measurably diverges from scoring-time behavior, and (ii) the auction exhibits the lowest 3-gram repetition of all systems (0.387–0.483 vs. best specialist 0.517–0.601; ensemble collapses to 0.78–0.83)—an implicit anti-repetition effect from bid competition.

A.7 Damage Curve: Layer-Tying Topology Search on Qwen3-1.7B

Prior to any uptraining, F25 measured the conversion damage when transforming Qwen3-1.7B (28 layers, 1.721B params, $d = 2048$, $d_{\text{ff}} = 6144$, 16 heads with 8 key-value heads) to the EqLMCore topology (explicit outer blocks around an iterative core). The sweep covered outer-layer counts ($n_{\text{pre}} = n_{\text{post}} \in \{6, 8\}$), core iteration counts ($n_{\text{cores}} \in \{1, 2, 4\}$), and initialization strategies (average layer-wise mean vs. per-layer stepwise mean). Perplexity is measured on a 3-passage general sample; baseline (untouched Qwen3-1.7B) is 6.01.

Table 14: F25 damage curve: perplexity under layer-tying topology variants, no uptraining. All configurations show severe damage pre-uptraining. Average initialization (elementwise mean of layer weights) preserves function 10–100× better than stepwise. Explicit outer layers are the primary driver of recovery; additional cores show diminishing returns.

Outer layers	Cores	Avg. init ppl	Stepwise init ppl	Param count	Param ratio to base
6+6	1	18465	2.17e5	1.027B	59.7%
6+6	2	19432	2.31e5	1.121B	65.2%
6+6	4	19734	2.54e5	1.310B	76.1%
8+8	1	1909	2.19e5	1.167B	67.8%
8+8	2	1933	2.28e5	1.262B	73.4%
8+8	4	2107	2.51e5	1.450B	84.2%
Baseline (no tying)	—	6.01	—	1.721B	100.0%

Three observations: (1) average initialization dominates stepwise by 10–100× at every configuration (e.g., 8+8 with 1 core: 1909 vs. 2.19e5 ppl), establishing that an element-wise mean of weights preserves more learned function than selecting a representative layer; (2) increasing explicit outer layers from 6+6 to 8+8 with 1 core improves perplexity 10× (18465 → 1909), while adding a second core at fixed 8+8 yields negligible improvement (1909 vs. 1933), indicating the first and last two layers carry irreplaceable function and the interior is far more redundant; (3) all configurations start heavily damaged pre-uptraining (300–3000× baseline). The operating point selected for uptraining is 8+8 outer layers with 1 core and average initialization: 1.167B parameters (67.8% of base), starting perplexity 1909. SPEC 0011's parameter-ratio gate was amended from $\leq 60\%$ to $\leq 70\%$ based on this evidence, before any uptraining was attempted: configurations at 56–59% parameter ratio start 3–10× more damaged and would not be recoverable within the planned token budget.

B Hyperparameters

B.1 121M Pretraining: Data and Optimizer

All pretraining experiments used the BabyLM strict-small data set (10M unique tokens from books, simple-Wikipedia, and Wikitext-2) as a streaming corpus, repeated until reaching the designated step count. Batch size was 32 sequences; sequence length was 128 tokens after tokenization. Gradient clipping was applied at norm 1.0. The vocabulary consists of 50257 BPE tokens (GPT-2 tokenizer). Token embeddings and output logits shared weights (tied embeddings). Positional embeddings were learned, one per sequence position.

The optimizer was AdamW with learning rate 3×10^{-4}, weight decay 0.01 (decoupled), $\beta_1 = 0.9$, $\beta_2 = 0.999$, epsilon 1×10^{-8}. For arm A3 (EqLM implicit) and B1 (anytime), the weight decay is applied via decoupled weight decay (parameter update $\theta \leftarrow \theta(1 - \eta\lambda_{\text{wd}})$ prior to gradient-based step), not folded into gradients, to preserve AdamW equivalence when $\tau = 0$ (verified in F11).

B.2 EqLM Architecture Details

The EqLM model (arms A3 and B1) is a causal language model solving

$$z^* = f_\theta(z^*, x)$$

where f is a weight-tied transformer block. The architecture specifications:

Table 15: EqLM architecture parameters for the 121M experiments. The block is solved by Anderson acceleration with 12-iteration budget and relative-tolerance gate at 10^{-3}. All iterates at training time supervise cross-entropy loss (anytime arm B1).

Parameter	Value
Hidden dimension (d)	1704
Number of heads	12
Dimension per head	142
Feed-forward dimension (d_{ff})	6807
Solver	Anderson acceleration
Anderson history size	5
Fixed-point map form	post-LN (Eq. 3)
Max iterations (budget)	12
Tolerance (relative)	1×10^{-3}
Activation function	GELU

The post-LN map has two layer normalizations: one after attention and MLP (feeding into the next input injection), and one after MLP (feeding into the output logits). Causal masking is applied inside multi-head attention. The output logit scaling is $1/\sqrt{d} = 1/\sqrt{1704} \approx 0.0243$. The attention and MLP projections use standard linear layers; no spectral normalization is applied in the baseline arms (A3, B1), though it is available in the codebase.

B.3 Anytime Training

Arm B1 (anytime-unrolled) unrolls the fixed-point map to 12 explicit applications and supervises cross-entropy loss at intermediate iterates. Supervision weights at iterates 4, 8, and 12 are 0.15, 0.30, and 1.0 respectively. These are the per-iterate loss weights; the total loss is $L_{\text{total}} = 0.15 \cdot \text{CE}(z_4) + 0.30 \cdot \text{CE}(z_8) + 1.0 \cdot \text{CE}(z_{12})$, with logits computed from the output head applied to each intermediate representation. This deep-supervision principle biases the model toward early convergence even though evaluation runs the fixed-point solver.

B.4 Solver-Aware Auxiliary Loss

Arm B2 (F24, B2 in Table 10) adds an auxiliary loss term $\lambda \cdot r_{\text{rel}}$ where $r_{\text{rel}} = \|f(z^*, x) - z^*\|/(\|z^*\| + \epsilon)$ is the relative residual computed once after the no-gradient fixed-point solve, and λ is the auxiliary loss weight (set to 0.01 in the reported run). The total loss is $L_{\text{total}} = L_{\text{CE}} + \lambda \cdot r_{\text{rel}}$.

B.5 Preference Optimization

The preference study (H3) fine-tuned the 121M checkpoints using the DPO loss:

$$L_{\text{DPO}} = -\log \sigma \left(\beta \left(\log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right),$$

where y_w is the preferred response, y_l is the dispreferred response, $\beta = 0.1$ is the DPO beta parameter, π_{ref} is the frozen base model, and σ is the logistic sigmoid. The base is the corresponding 121M checkpoint from F20 (either explicit A1 or EqLM A3). Learning rate was 1×10^{-5}, batch size 8 preference pairs, 3 epochs, with the identical optimizer (AdamW, weight decay 0.01). For magnetic arms, MagneticAdamW applies $y_{\text{ref}} =$ frozen base weights with strength $\tau \in \{1 \times 10^{-3}, 1 \times 10^{-2}, 0.1, 1, 10\}$ depending on the arm. The DPO arm (P1, $\tau = 0$) is mathematically standard decoupled AdamW, verified for equivalence against torch.optim.AdamW on sparse-gradient parameters.

B.6 Specialist Models and Auction Setup

Two 30M-parameter specialists were trained from scratch on disjoint subsets of BabyLM strict-small: specialist S_A on child-directed speech (CHILDES), specialist S_B on simple-Wikipedia. Each specialist is a standard GPT-2-class decoder (12 layers, $d = 384$, 6 heads, $d_{\text{ff}} = 1536$) trained for 3000 steps on its domain subset with batch size 8, learning rate 3×10^{-4}, and a line-level 5% held-out evaluation split. The auction mechanism selects the specialist with the highest predicted probability for the next token at each position, and that specialist's distribution scores the true token. The winner pays the second-highest predicted probability as the second-price payment, implementing a truthful auction (F6). Confidence bids are target-independent (the specialist does not observe the true next token before bidding).

B.7 Qwen3 Conversion (F25)

Qwen3-1.7B configuration: 28 transformer layers, hidden dimension $d = 2048$, feed-forward dimension $d_{\text{ff}} = 6144$, 16 attention heads with 8 key-value heads (multi-query attention), head dimension 128, vocabulary size 151936, tied embeddings. The conversion to the EqLMCore topology splits the 28 layers into n_{pre} explicit pre-layers, n_{cores} iterative core layers (each identical, weight-tied), and n_{post} explicit post-layers, with $n_{\text{pre}} = n_{\text{post}}$ by design. Initialization strategies tested: (i) average: each core layer's weights are the element-wise mean of the corresponding layers from the original 28-layer stack, and (ii) stepwise: each core layer picks a representative layer from its assigned range (e.g., if $n_{\text{pre}} = n_{\text{post}} = 8$ and $n_{\text{cores}} = 1$, the core is initialized from layer 14). Perplexity is computed on a 3-passage held-out sample (not the full 10M-token stream), using the base 6.01 as reference. Uptraining results are in progress and are not reported in this appendix.